



西安交通大学
XI'AN JIAOTONG UNIVERSITY

实验三 射频功率分配/合成器 设计、仿真与测试



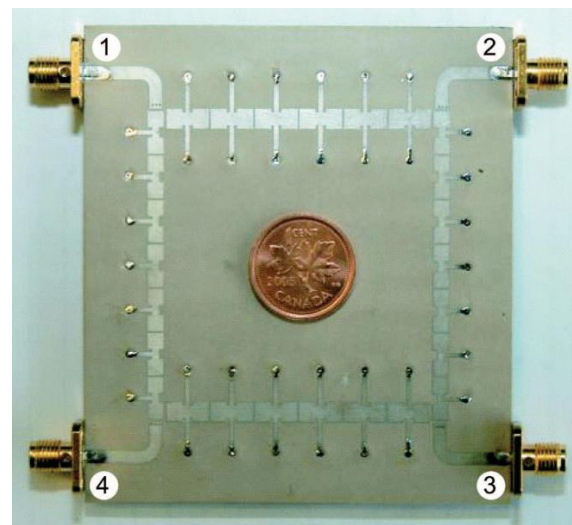
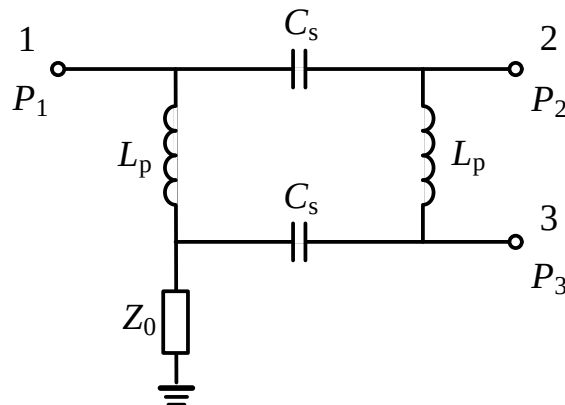
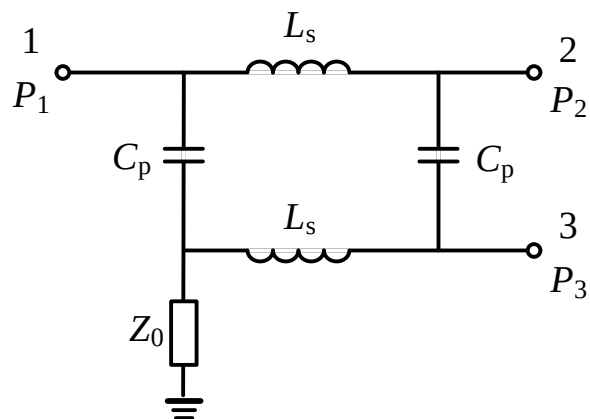
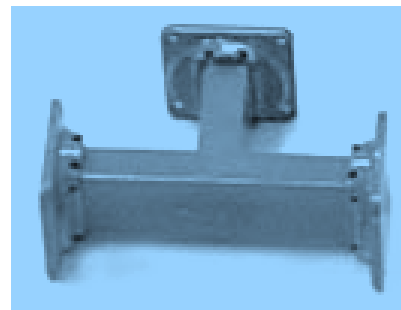
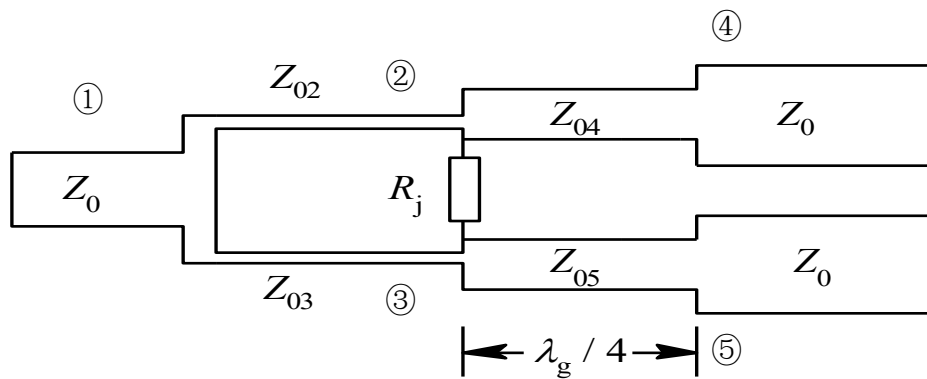
一、实验目的

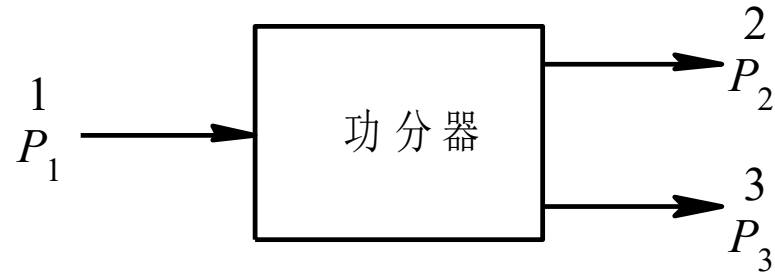
1. 了解功率分配器的原理及基本设计方法
2. 掌握威尔金森功分器的结构、工作原理及S参量
3. 了解利用ADS进行电路优化仿真的基本步骤及方法
4. 掌握利用ADS微带线计算工具LinCalc计算、设计微带线
5. 了解利用ADS在电路板级进行电路仿真的方法与步骤。



二、基本理论

将一路微波功率按一定比例分成 n 路输出的功率元件称为功率分配器。按输出功率比例不同,可分为等功率分配器和不等功率分配器。





技术指标:

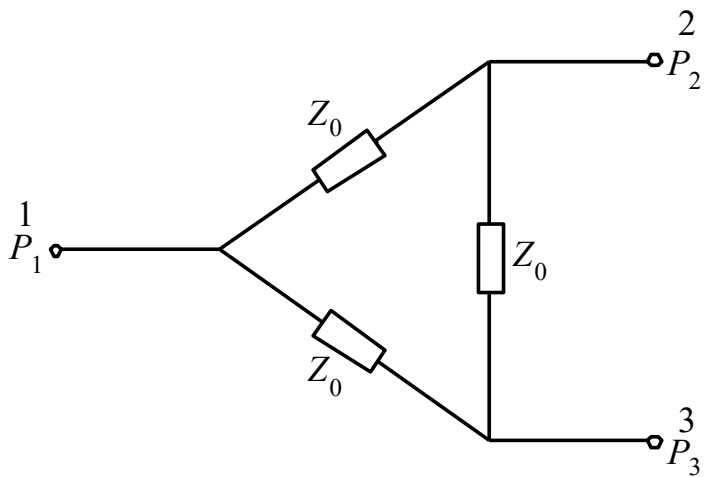
- **频率范围:** 分配器的工作频率
- **承受功率:** 分配器/合成器所能承受的最大功率
- **功率分配比:** 主路到支路的功率分配比
- **插入损耗:** 输入输出间由于传输线（如微带线）的介质或导体不理想等因素,考虑输入端的驻波比所带来的损耗
- **驻波比:** 每个端口的电压驻波比
- **隔离度:** 支路端口间的隔离程度



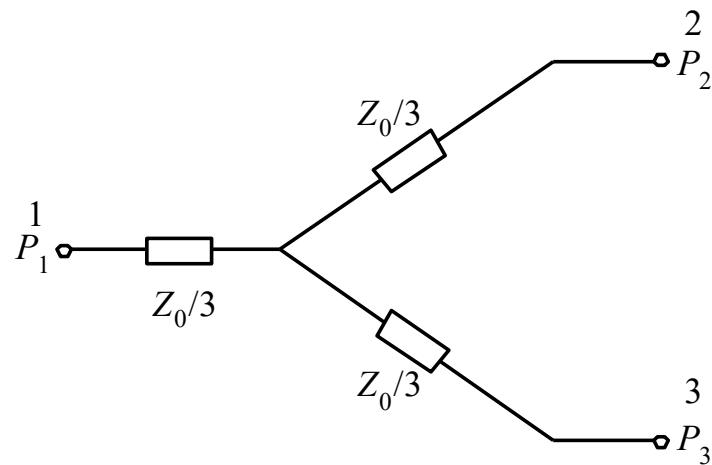


2.1 集总参数功率分配器

1. 电阻式（等功率）



(a)

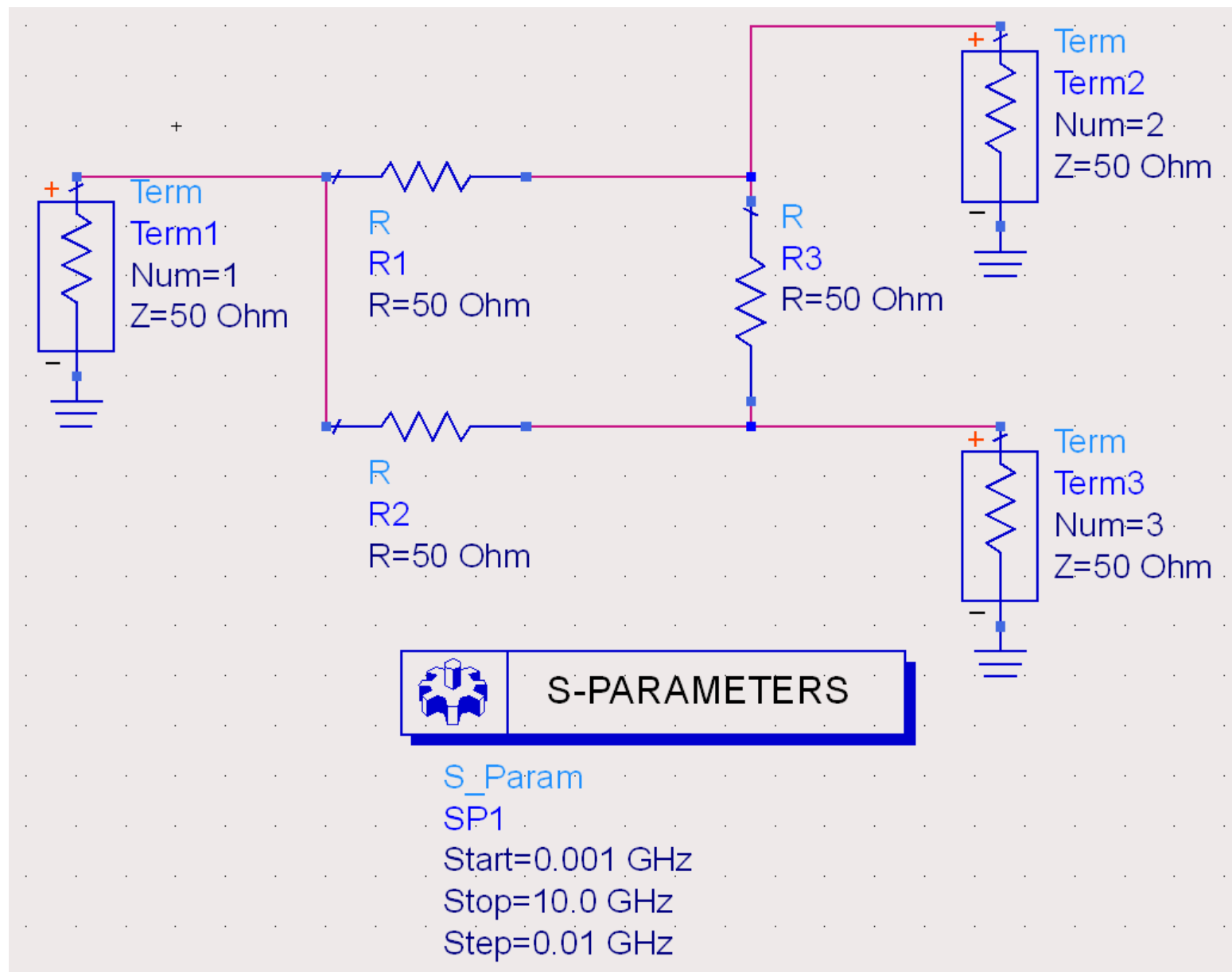


(b)

△形和Y形电阻式功率分配器

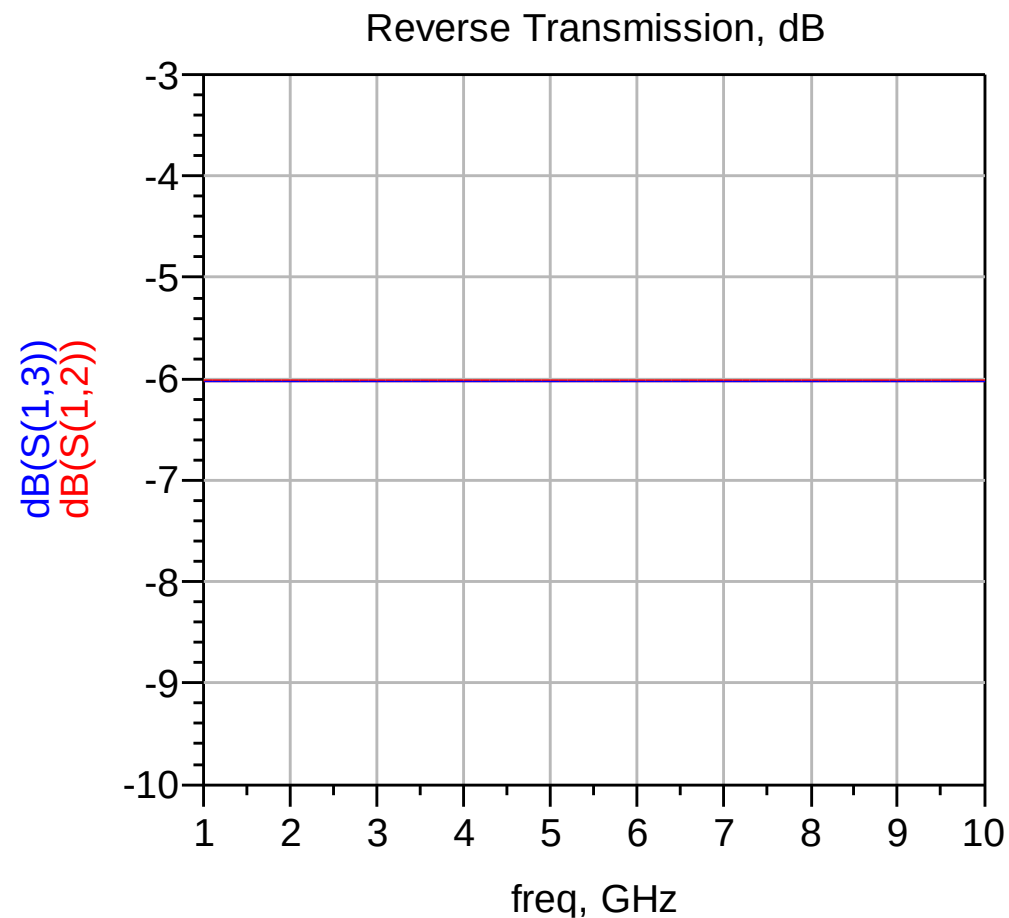


ADS仿真：△形电阻式功率分配器





ADS仿真：△形电阻式功率分配器



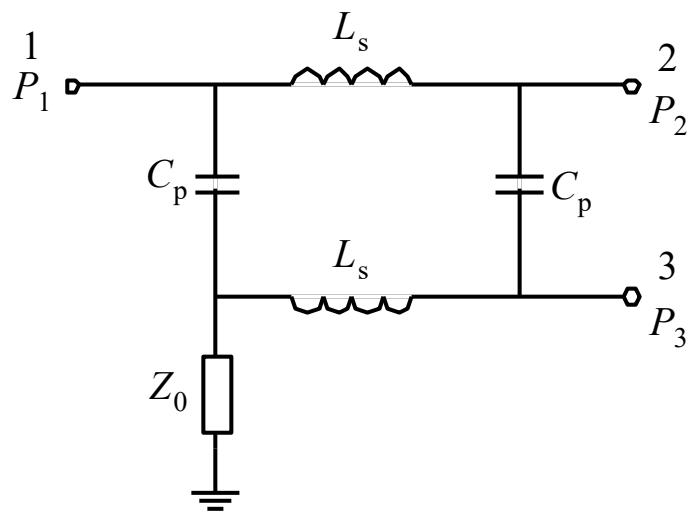
优点：频宽大, 布线面积小, 设计简单

缺点：功率衰减较大



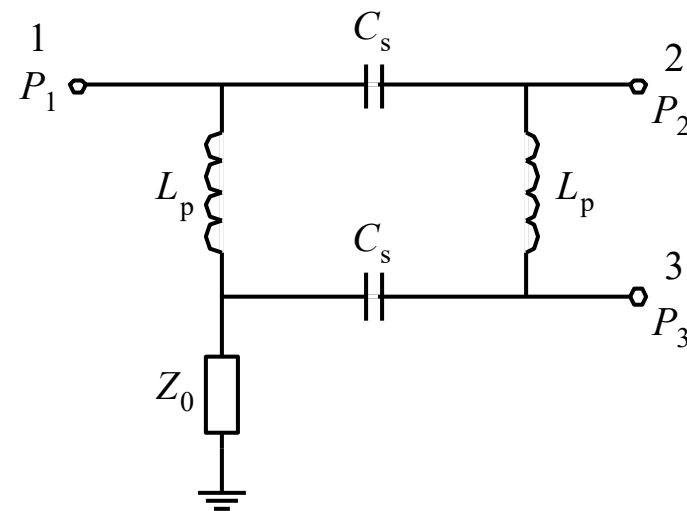


2. 集总L-C式低通型和高通型功率分配器



(a)

$$L_s = \frac{Z_0}{\sqrt{2}\omega_0}$$
$$C_p = \frac{1}{\omega_0 Z_0}$$
$$\omega_0 = 2\pi f_0$$

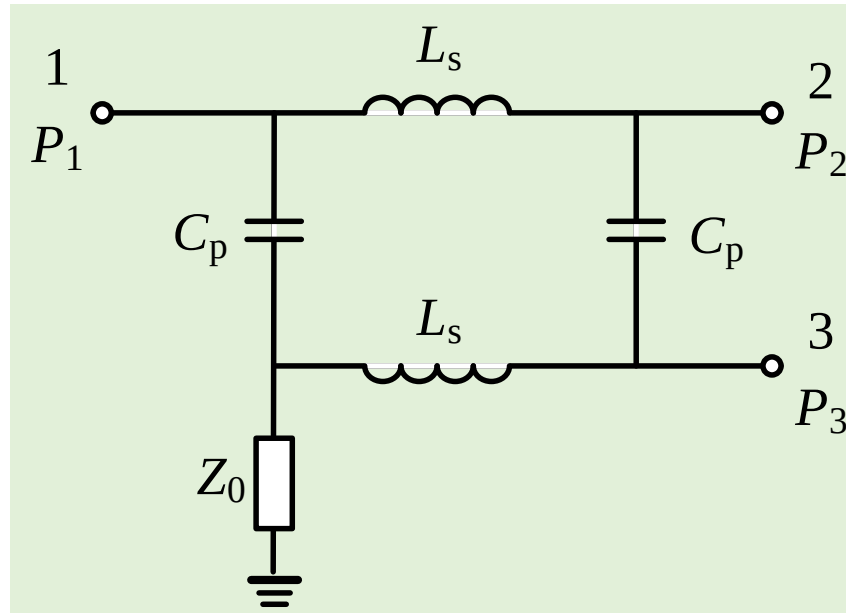


(b)

$$L_p = \frac{Z_0}{\omega_0}$$
$$C_s = \frac{\sqrt{2}}{\omega_0 Z_0}$$
$$\omega_0 = 2\pi f_0$$



ADS仿真：中心频率为2.4GHz的集总参数 L-C 式 低通型功率分配器，传输线特征阻抗为50欧姆。



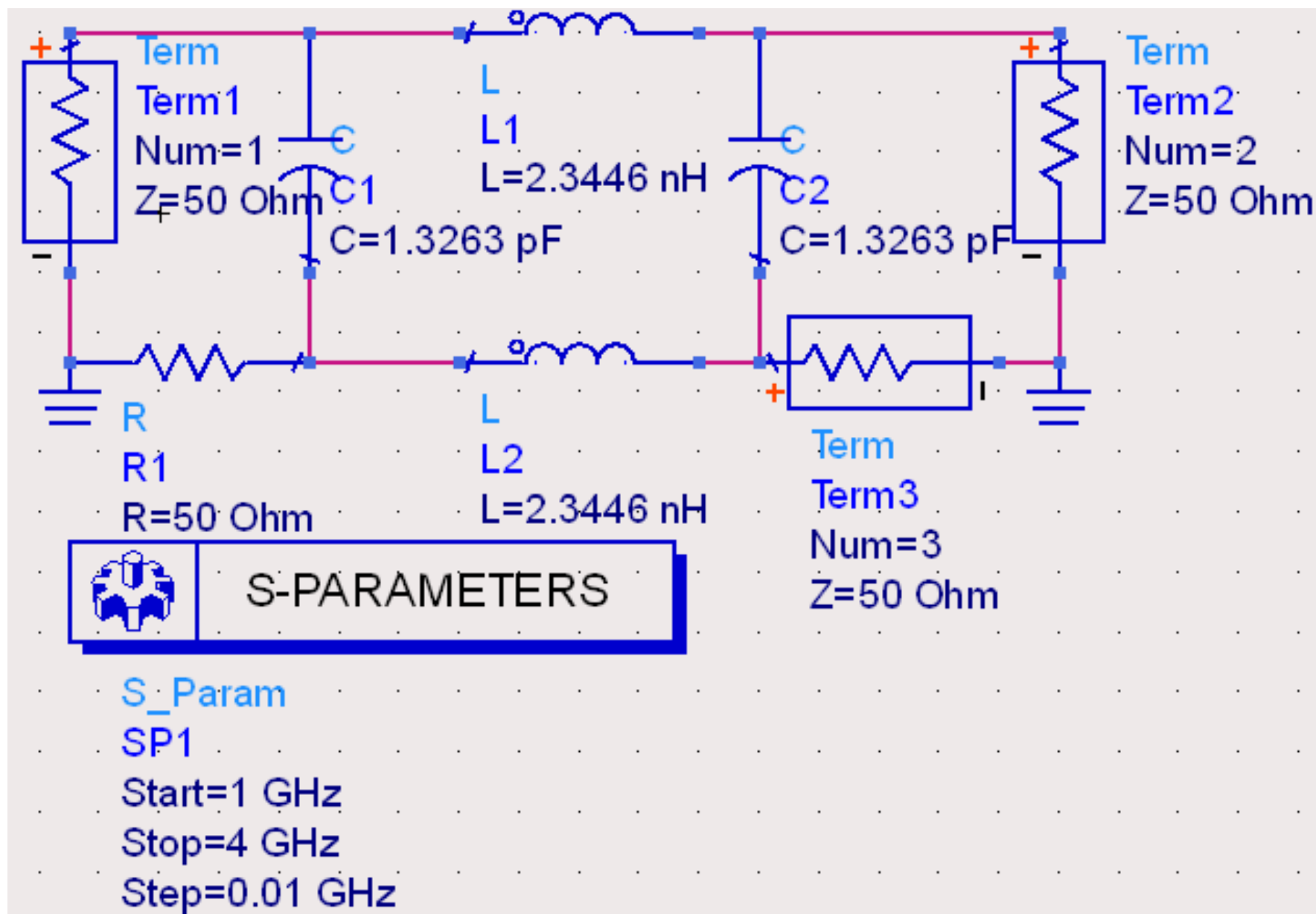
$$L_s = \frac{Z_0}{\sqrt{2}\omega_0} = \frac{50}{\sqrt{2} \times 2 \times \pi \times 2.4 \times 10^9} = 2.3446 \times 10^{-9} = 2.3446 \text{ nH}$$

$$C_p = \frac{1}{\omega_0 Z_0} = 1.3263 \times 10^{-12} = 1.3263 \text{ pF}$$



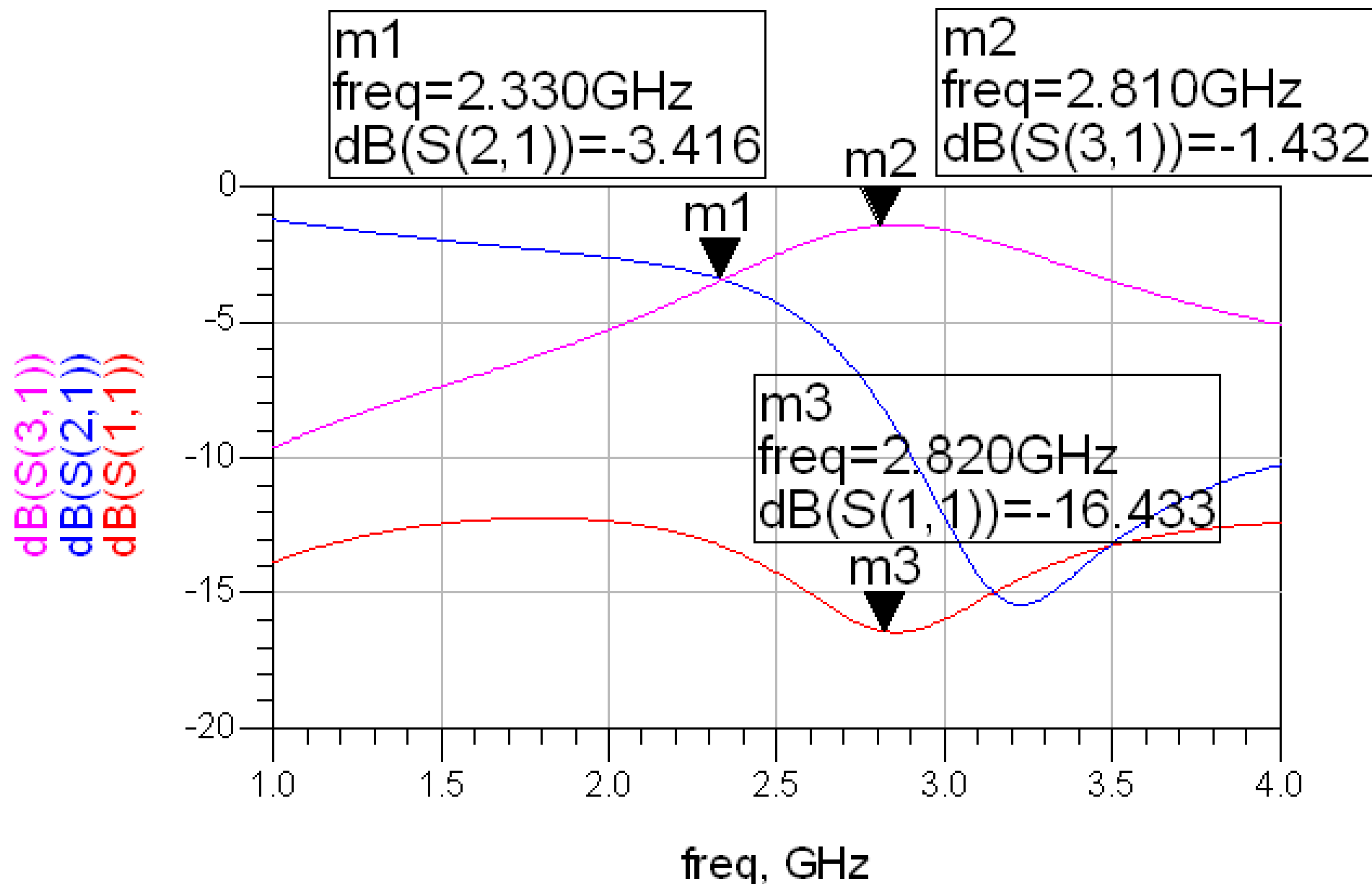


ADS仿真：L-C 式低通型功率分配器



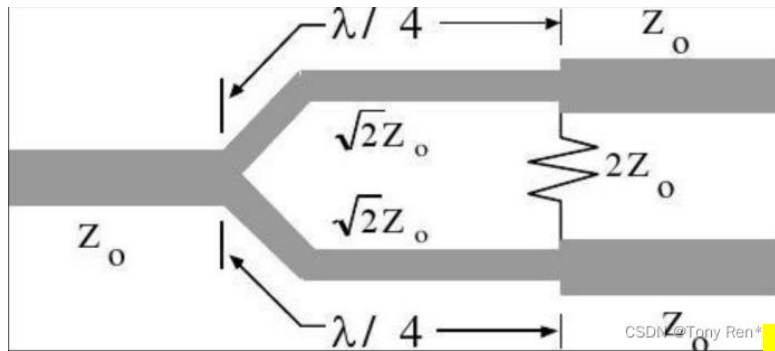


ADS仿真：L-C 式低通型功率分配器

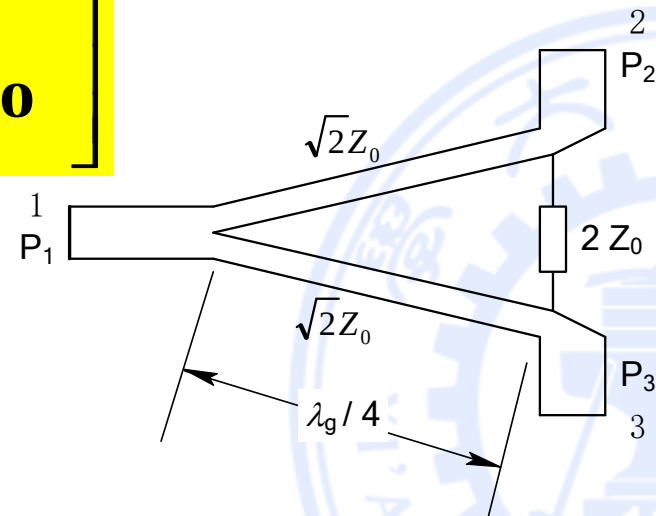
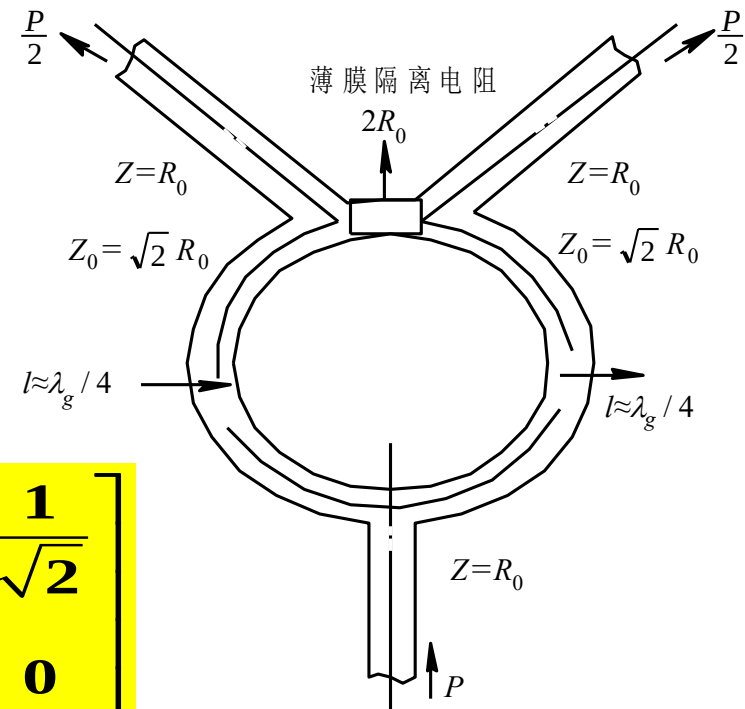
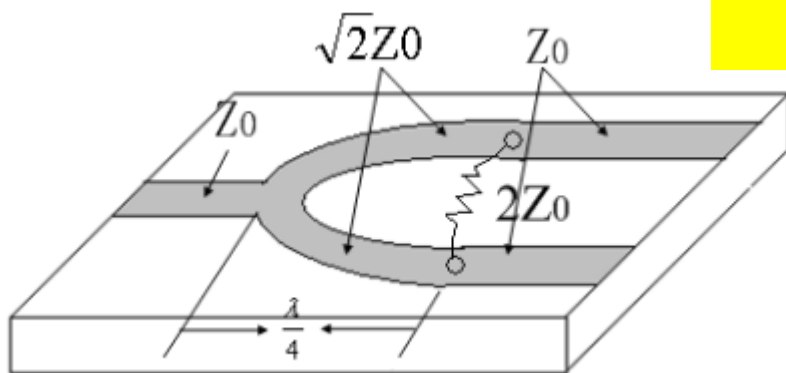




2.2 威尔金森功分器



$$[s] = -j \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & 0 & 0 \end{bmatrix}$$





威尔金森功分器的设计与仿真

设计指标:

频率范围: 0.9-1.1GHz

频带内输入端口的回波损耗: $S_{11} < -20\text{dB}$

频带内插入损耗: $S_{21} > -3.1\text{dB}$, $S_{31} > -3.1\text{dB}$

隔离度: $S_{32} < -20\text{dB}$

微带板材参数:

H: 基板厚度(0.8 mm)

ϵ_r : 基板相对介电常数(4.3)

μ_r : 磁导率(1)

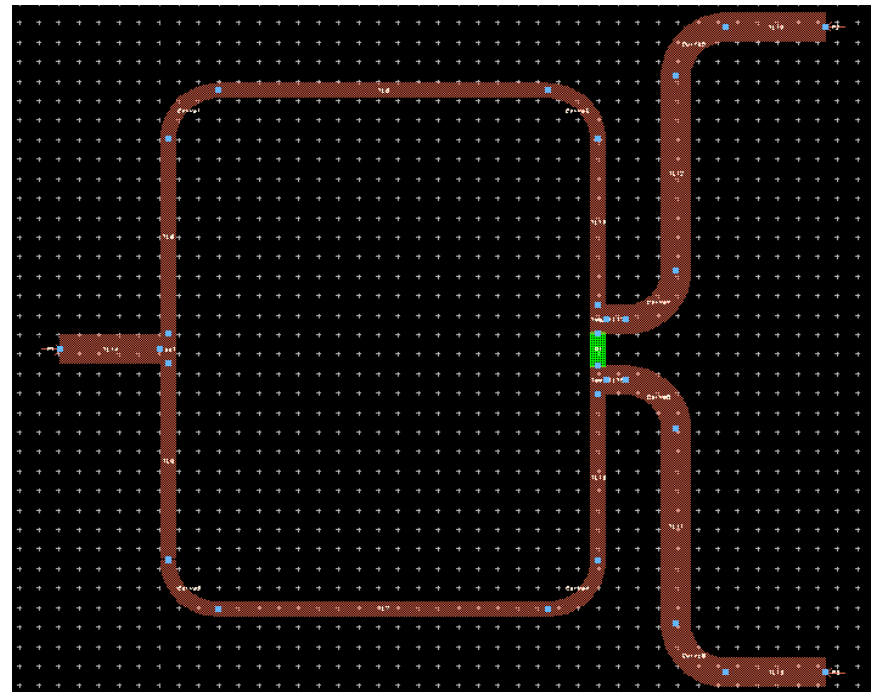
Cond: 金属电导率($5.88\text{E}+7$)

Hu: 封装高度($1.0\text{e}+33$ mm)

T: 金属层厚度(0.03 mm)

TanD: 损耗角正切($1\text{e}-4$)

Rough: 表面粗糙度(0 mm)





1. 创建新的WorkSpace

Standard ADS Layers, 0.0001 millimeter layout resolution

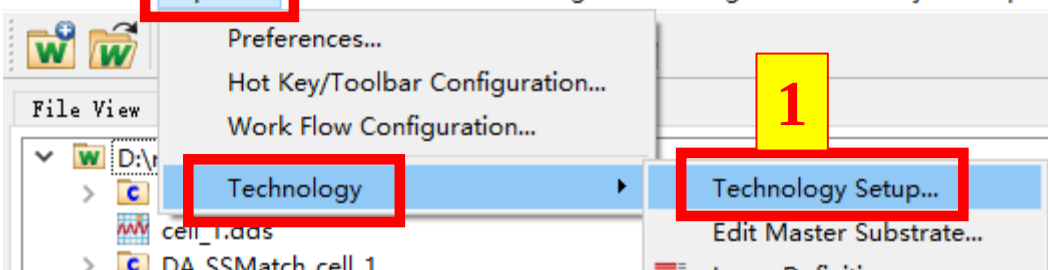




如果是高版本软件

Advanced Design System 2022 (Main)

File View Options Tools Window DesignKits DesignGuide CoilSys Help



Technology Setup

View Technology for this Library: MyLibrary@_lib

Referenced Libraries Layout Units

USE CAUTION when changing the units or resolution of a library that has existing layout views. This can lead to serious problems in your layouts.

Database Resolution

☒ Enable Database Resolution

1000

Units

☒ Enable Units

☐ nanometer

☐ micron

☒ millimeter

☐ centimeter

6

Manufacturing Grid

Technology Setup

View Technology for this Library: MyLibrary@_lib

Referenced Libraries Layout Units

2

Referenced Libraries

ads_schematic_layers

Referenced Libraries

ads_standard_layers

ads_schematic_layers

5

Show Advanced Options

3

Add Referenced Library ...

Add Referenced Library

Select a library

Library Name	Layout Unit	Layout Resolution	Layout Layers
ads_standard_layers			default, cond, cond2, resi, diel, diel2, hole, bond,...
ads_standard_layers_ic			default, cond, cond2, resi, diel, hole, bond, symb...
empro_standard_layers			empro_3dc_pins, empro_3dc_artwork

4



➤ 建立New Schematic

The screenshot displays a software interface for creating a new schematic. On the left, a library of components is shown, with the title bar "Lines-Microstrip" circled in red. The components include:

- MSUB (Microstrip Substrate)
- MSUBST (Microstrip Substrate with 3 layers)
- Maclin (Microstrip Line)
- Maclin3 (Microstrip Line with 3 segments)
- MSABND (Microstrip Bend)
- MSOBND (Microstrip Bend with 3 segments)

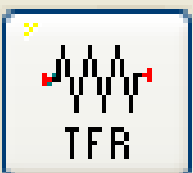
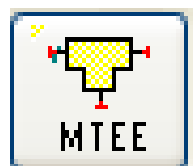
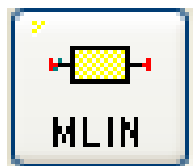
On the right, the properties for the selected "MSub" component are listed:

- MSUB
- MSub1
- H=0.8 mm
- Er=4.3
- Cond=5.88E+7
- T=0.03 mm
- TanD=0.0001



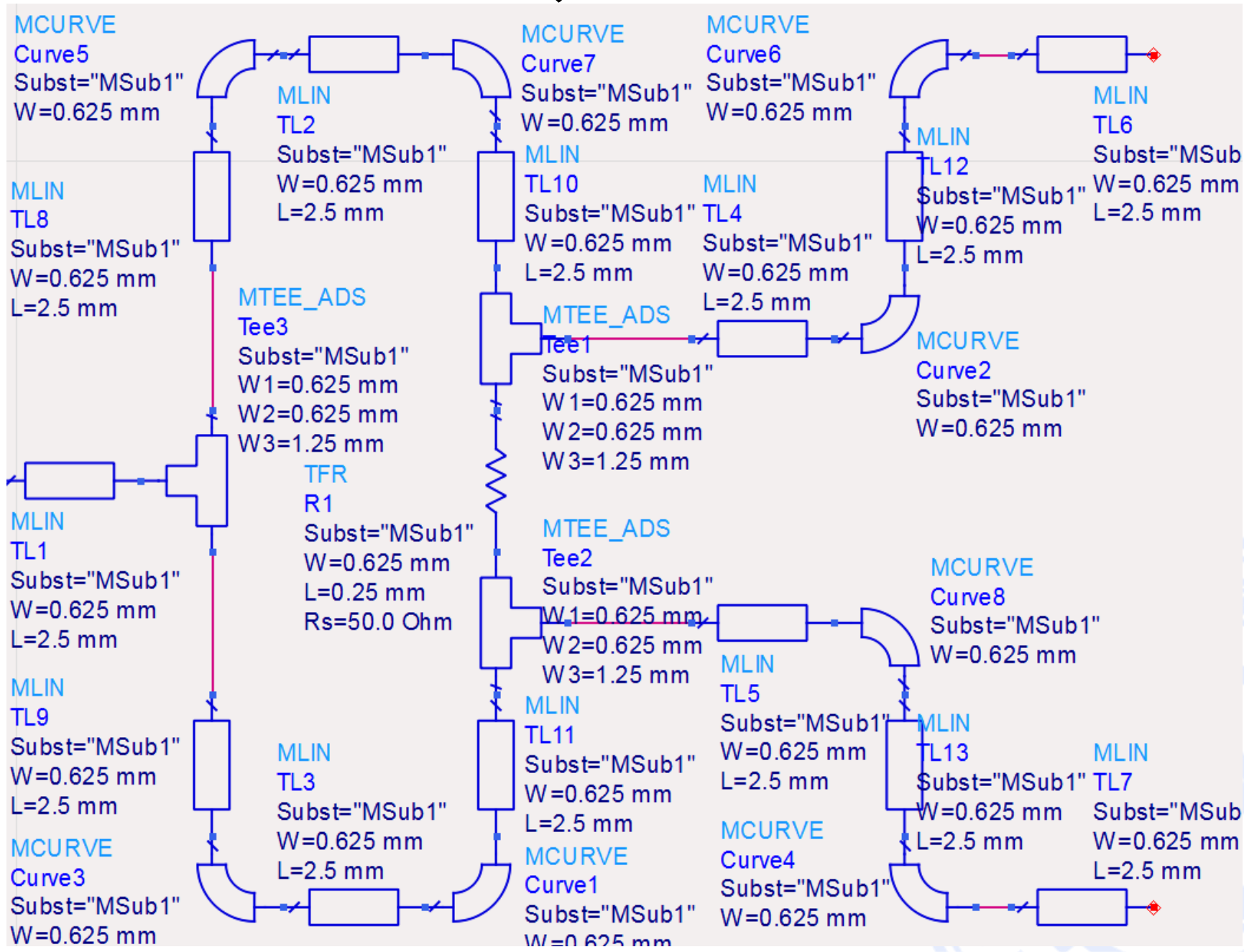


2. 画原理图，选择、放置微带线控件，并用线连接



Ctrl+R 旋转

F5 移动文字





3. 利用微带线计算工具计算微带线尺寸参数

➤ 执行菜单命令【Tools】-【LineCalc】-【Start LineCalc】

➤ 修改参数（单位改为mm），

然后单击



按钮就可以算出微带线的线宽1.52 mm

Substrate Parameters

ID MSUB_DEFAULT

Er	4.300	N/A	1
Mur	1.000	N/A	
H	0.800	mm	
Hu	3.9e+34	mm	
T	0.030	mm	
Cond	5.88e7	N/A	
TanD	1.000e-4	N/A	

Component Parameters

Freq 1.000 GHz

Physical

W 1.521330 mm

L 106.568000 mm

Synthesize 5

Analyze

Electrical

Z0 50.000 Ohm

E_Eff 230.000 deg

4 单位改为mm

3



- Z_0 项改入70.7 Ohm, 在E_Eff中输入90deg (对应四分之一波长), 重新计算得微带线的线宽0.7889 mm, 线长为42.8971mm

ID		
MSUB_DEFAULT		
Er	4.300	N/A
Mur	1.000	N/A
H	0.800	mm
Hu	3.9e+34	mm
T	0.030	mm
Cond	5.88e7	N/A
TanD	1.0e-4	N/A
Rough	0.000	mm
		N/A

Physical		
W	0.788886	mm
L	42.897100	mm
		N/A
		N/A

Synthesize	Analyze
	

Electrical		
Z0	70.700	Ohm
E_Eff	90.000	deg



4. 设置优化变量

➤ 设置优化参数: 在原理图中插入 “VAR” 控件



➤ 双击 “VAR”控件,
设置w1、w2、L 三个变量
(此处不设单位)

w1=1.52

w2=0.79

L=16

Cell name: VAR
View name: symbol
Instance name: VAR1

Swap Component...

Select Parameter

- w1=1.52
- w2=0.79
- L1=16**

Variable or Equation Entry Mode

Standard

Name: L1

Variable Value: 16

None

Equation Editor...

Tune/Opt/Stat/DOE Setup...

☒ Display parameter on schematic

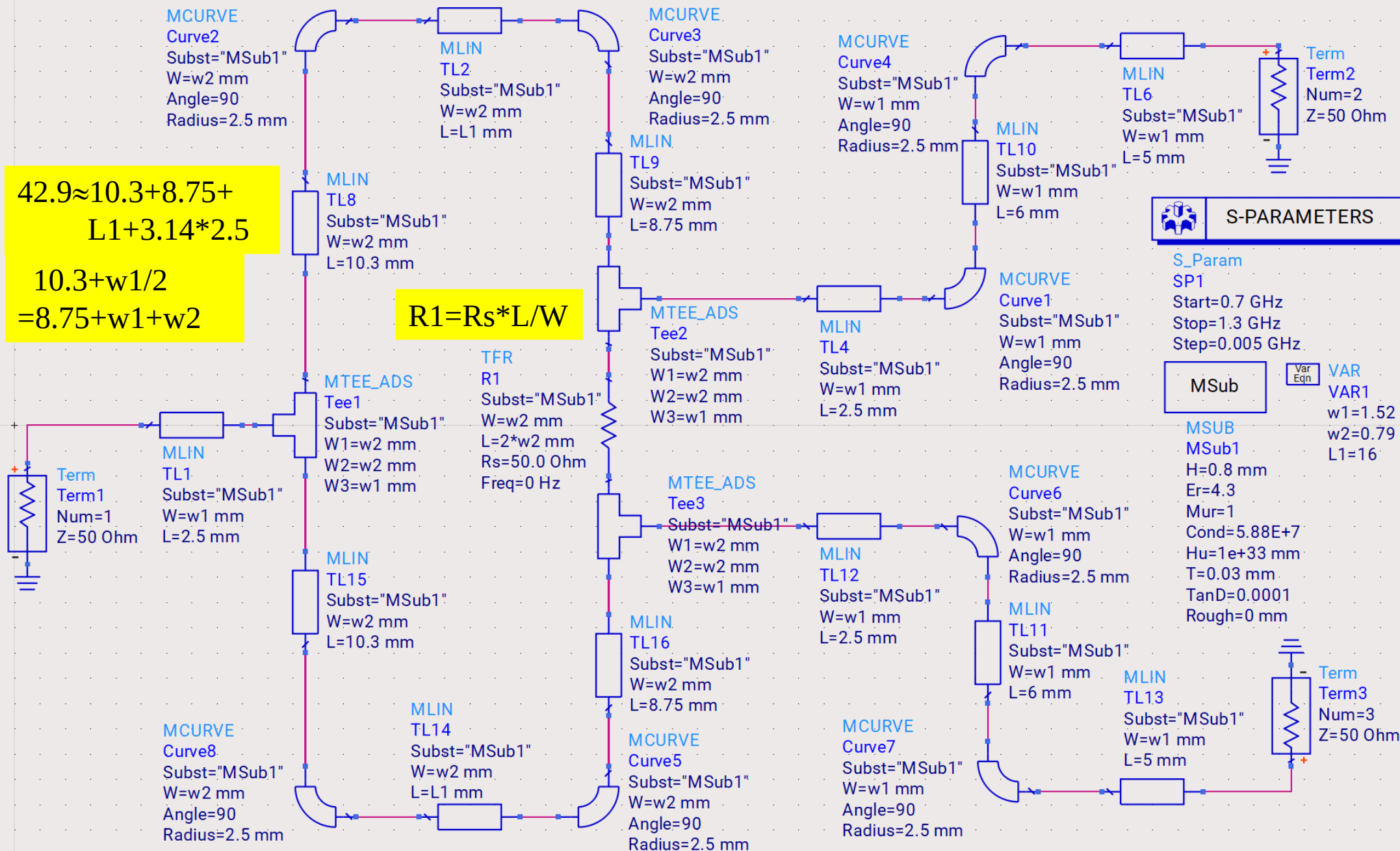
Component Options... Reset

Variable Value: Variable equation

OK Apply Cancel Help

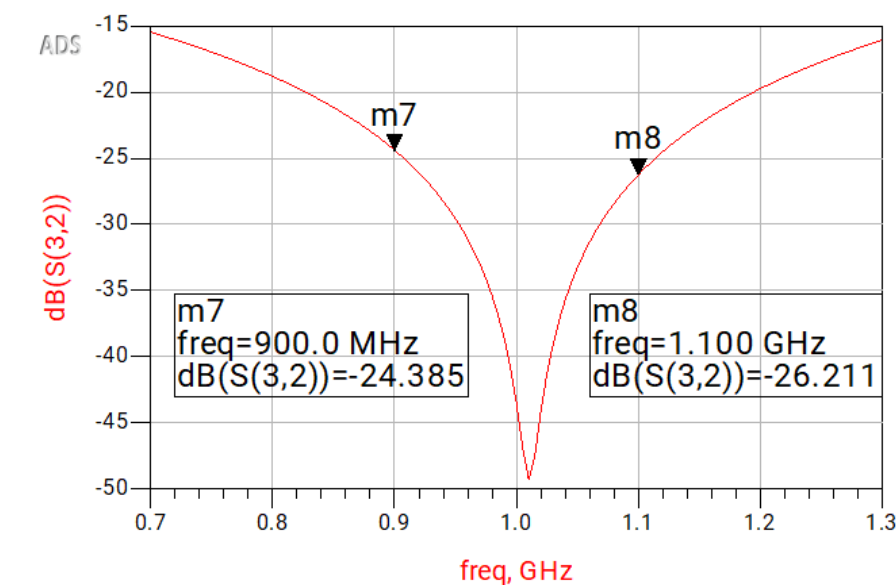
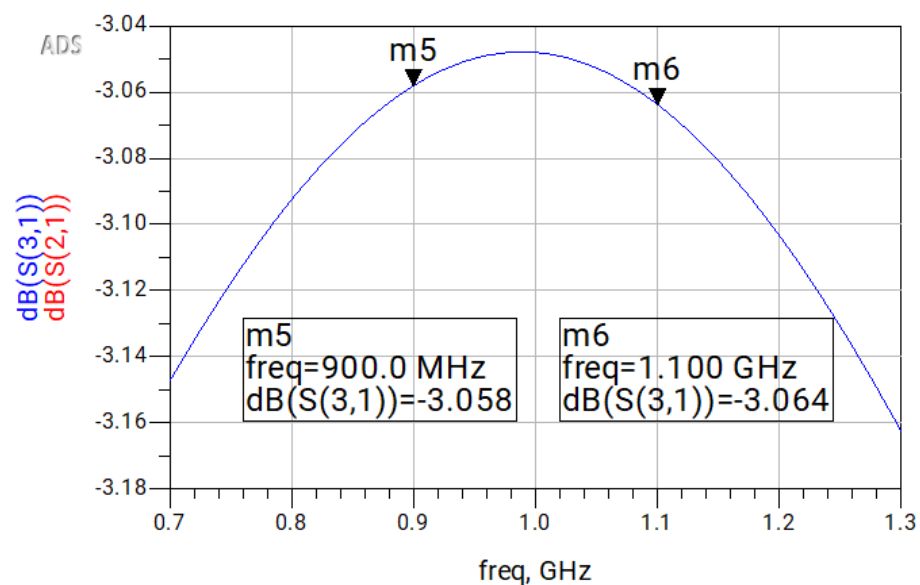
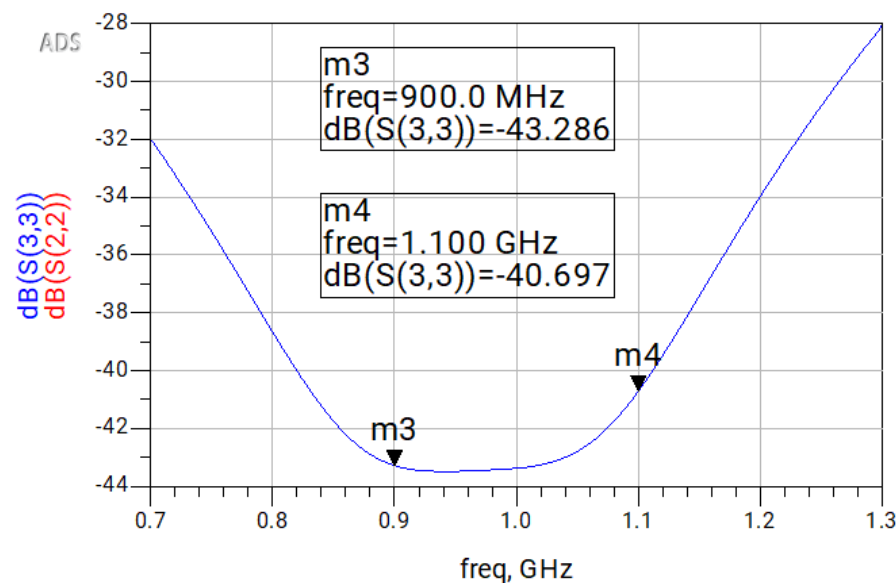
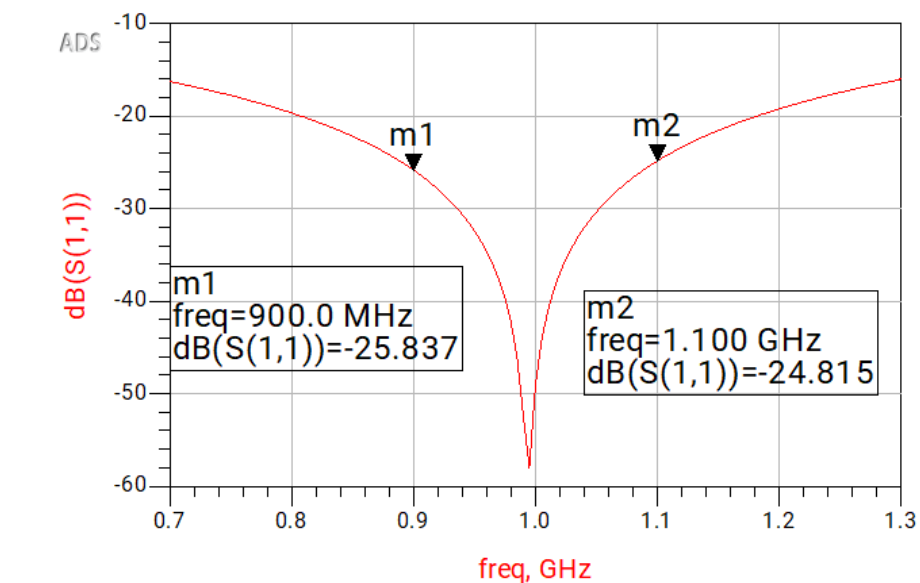


➤ 设置微带线宽度W和长度L，加SP控件、端口





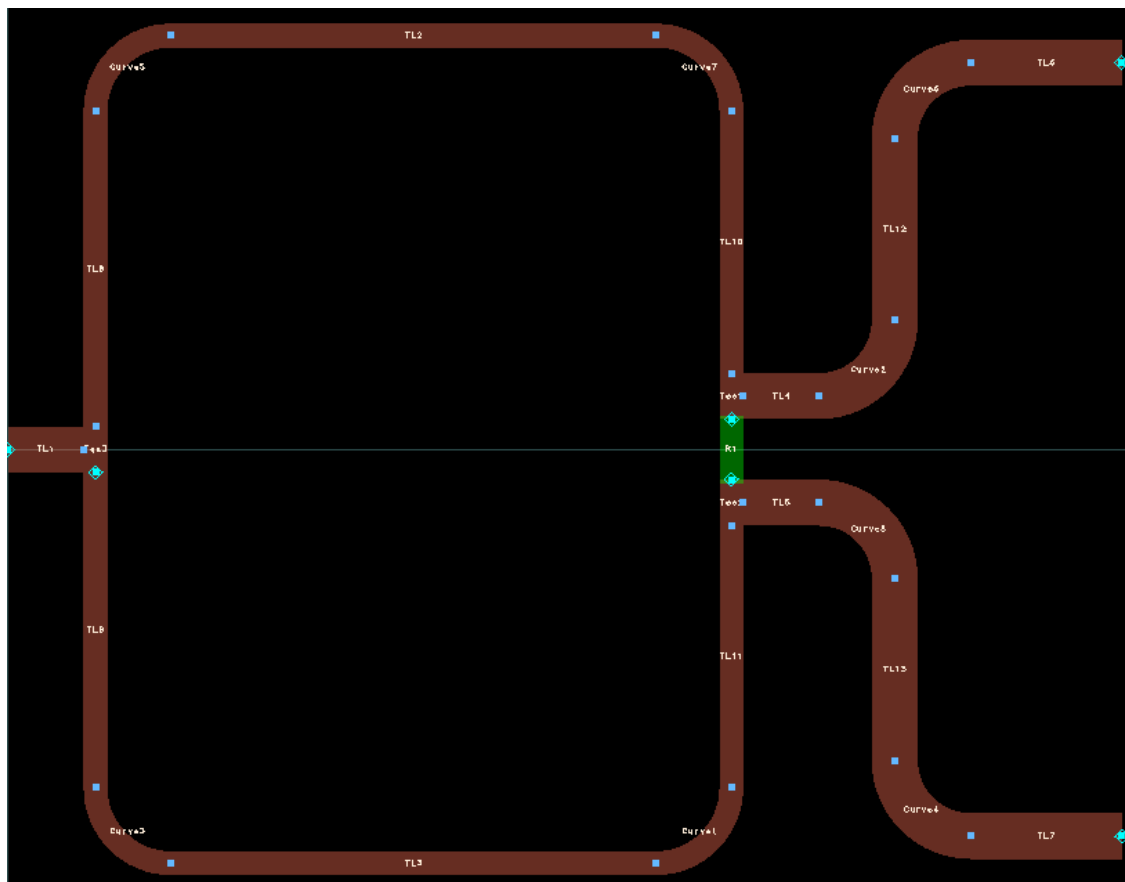
5. 原理图仿真结果





6. 功分器的版图生成

➤ 菜单命令【Layout】→【Generate/Update Layout】，单击OK





7. 功分器的版图仿真（设置电路板参数）

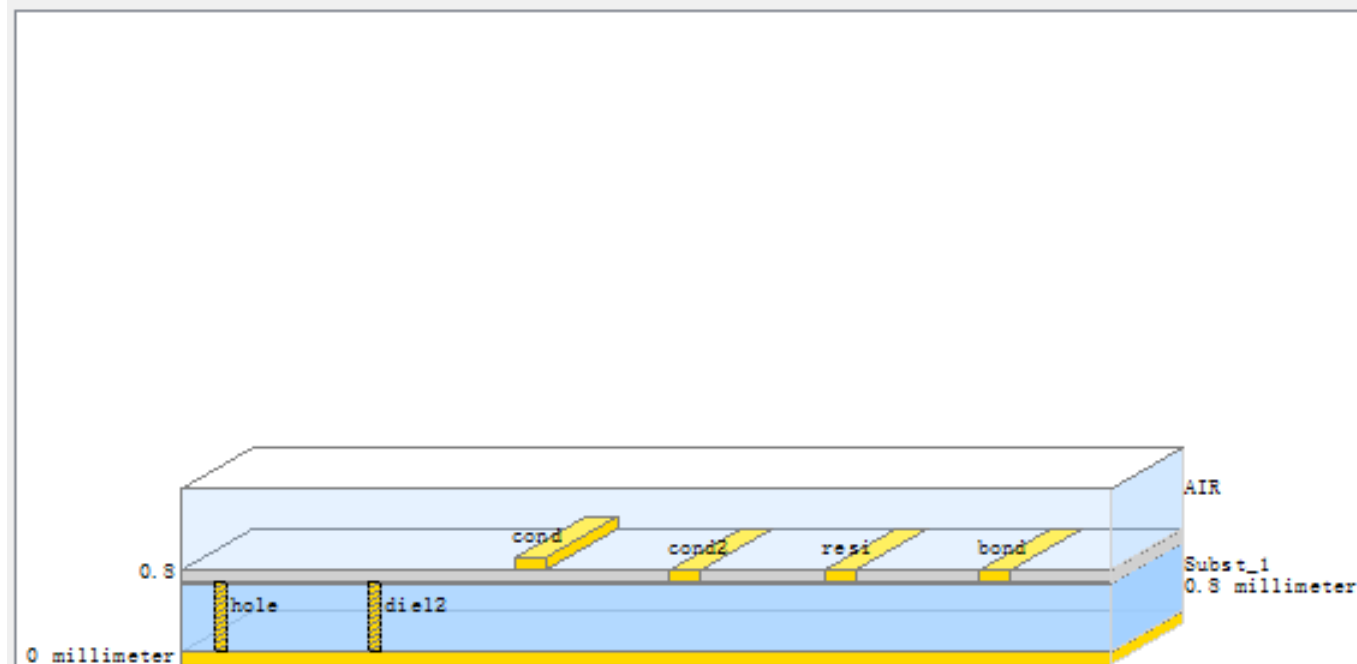
- 在版图窗口执行菜单命令【EM】→【Substrate】，全部单击【OK】，弹出“Substrate”设置窗口。
- 【File】→【Import】→【Substrate From Schematic】，选择相对应的原理图，更新板材参数。

cell_1 [Power_lib] (Substrate):7

File Technology View Options Tools Window Help



Substrate Name: cell_1



Use right mouse context menus to:

Select items on the substrate and

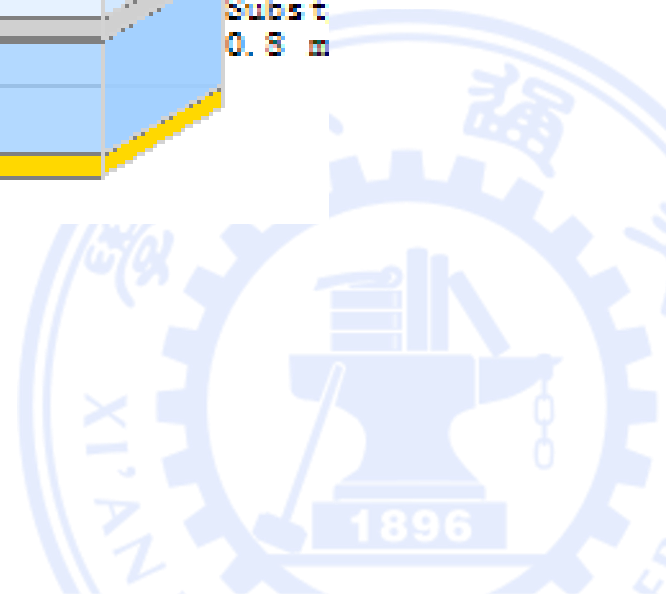
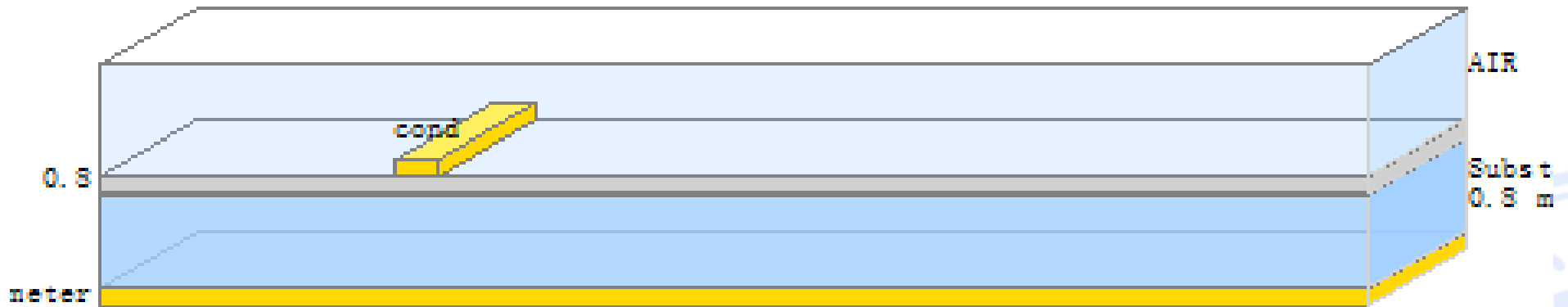
Entire Substrate

Bounding area layer: <none>

Select a substrate



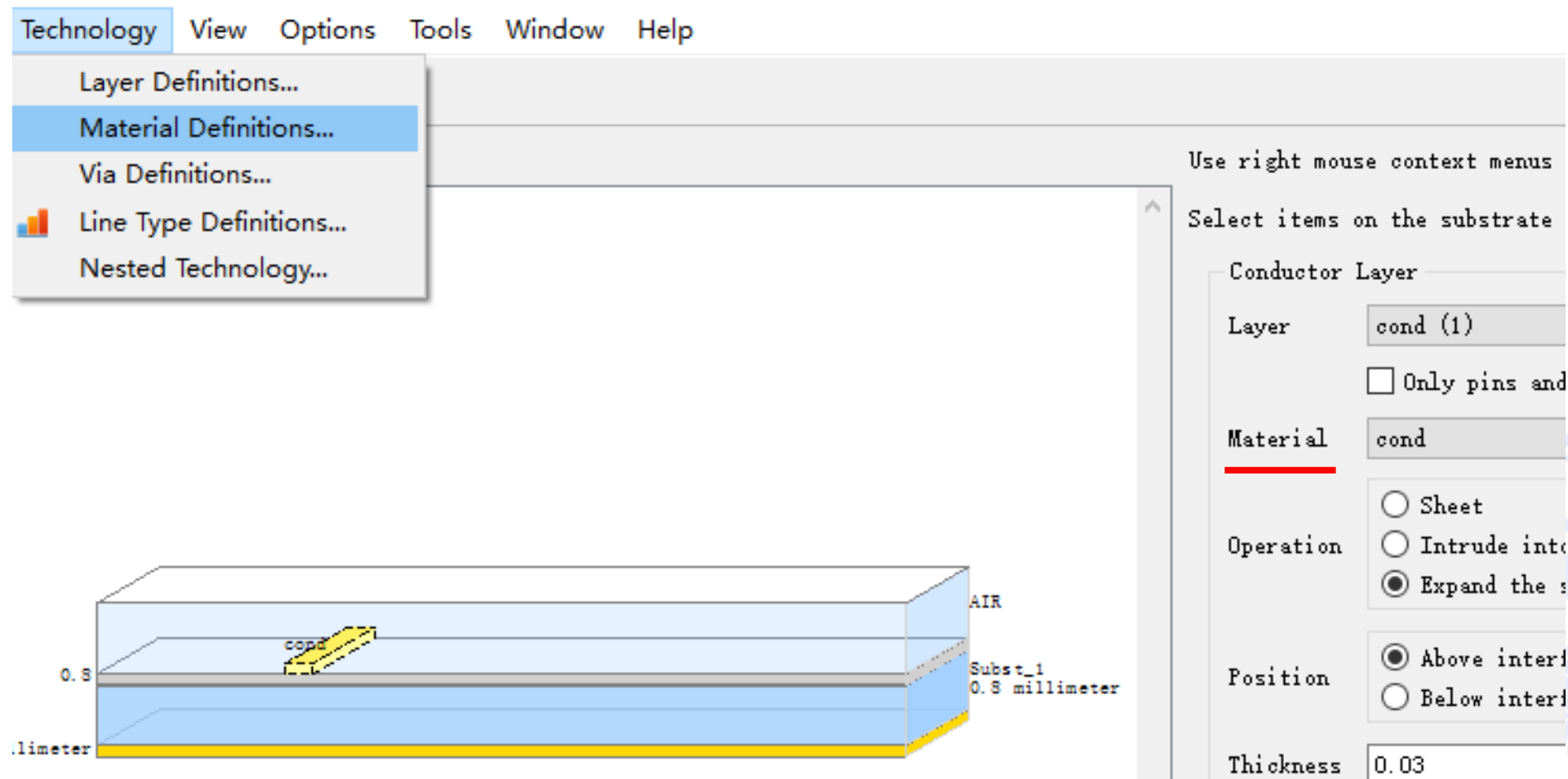
- 鼠标左键单击选中不需要的导体，再单击右键，弹出【Unmap】，单击之，再点【OK】，去掉该导体等
- 单击工具栏里面的保存按钮 





➤ 修改(检查)参数【Technology】→【Material Definitions】

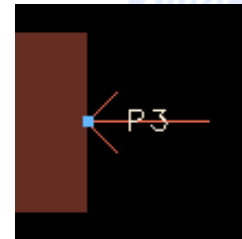
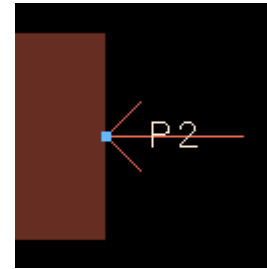
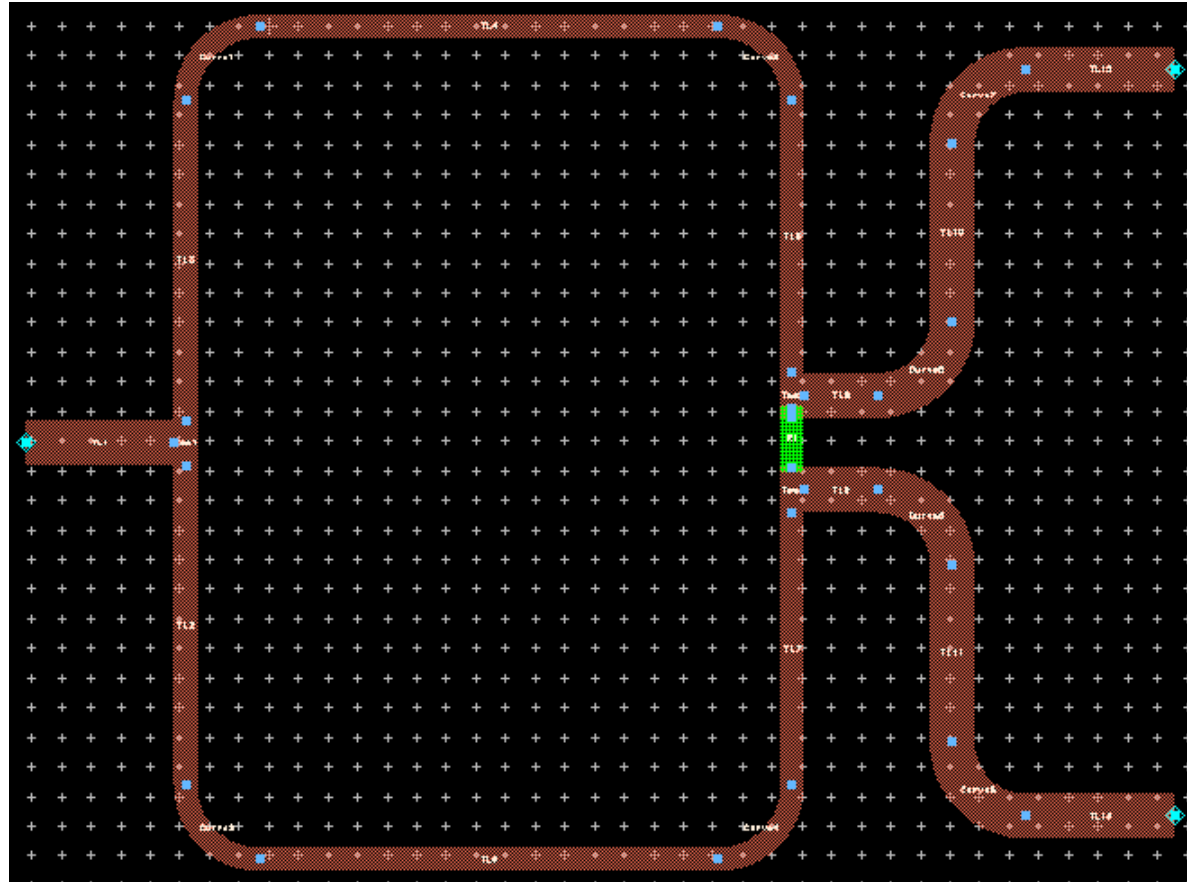
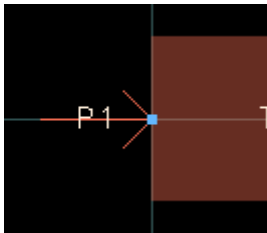
➤ 或者选中想要修改的材料，然后在右侧修改“Material”的参数





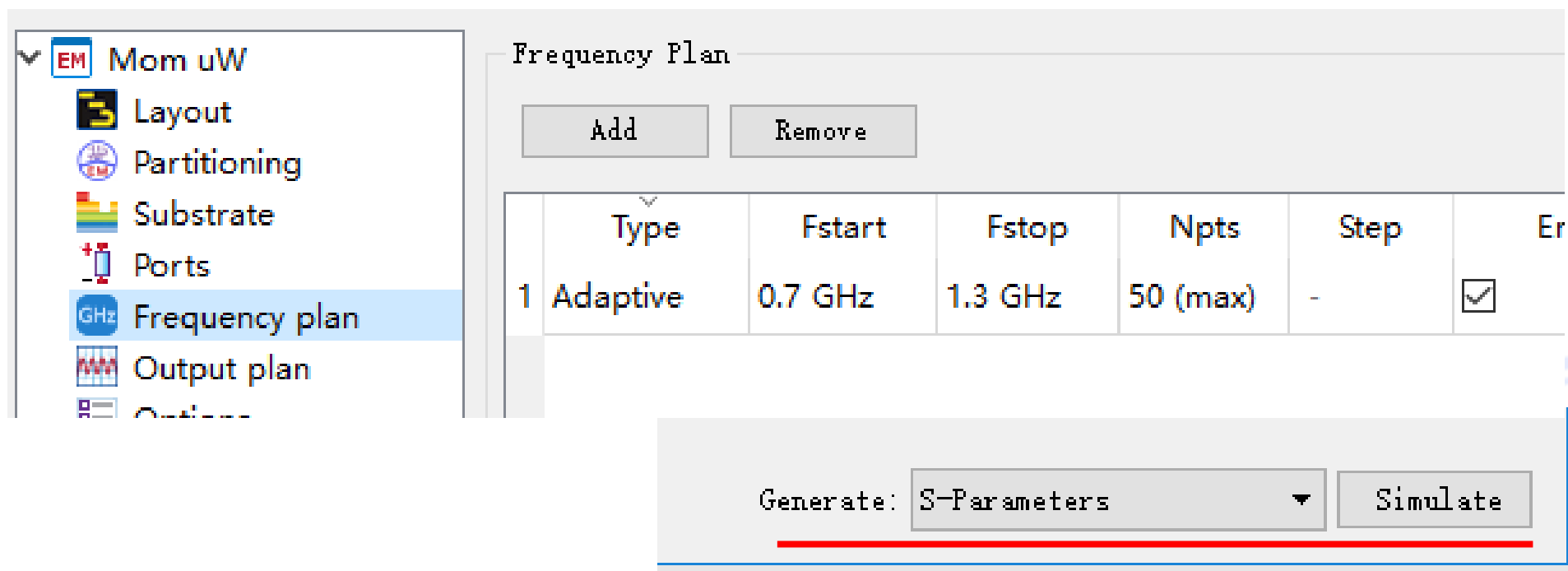
8. 功分器的版图仿真(添加端口及仿真参数设置)

- 执行菜单命令【Insert】→【Pin】或单击 ，插入三个端口分别于端口1、端口2、端口3





- 菜单【EM】→【Simulation Setup】或者 ，打开仿真设置窗口
- 选中【Substrate】和【Ports】检查设置是否正确
- 选中【Frequency plan】设置仿真参数。然后仿真。

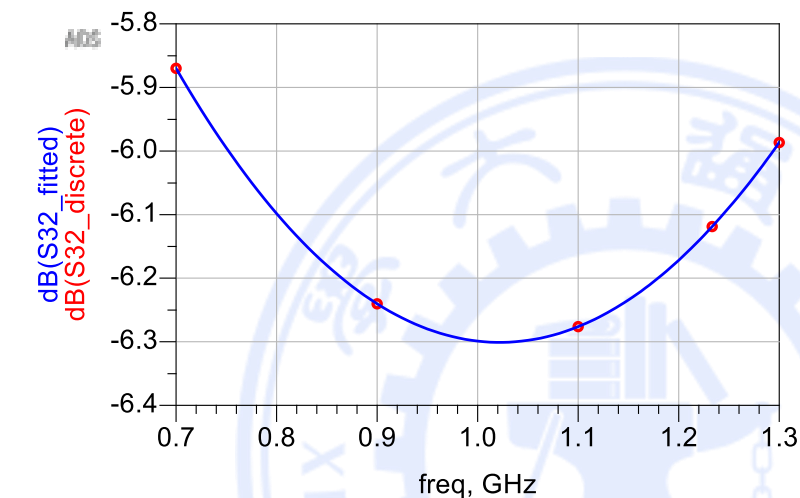
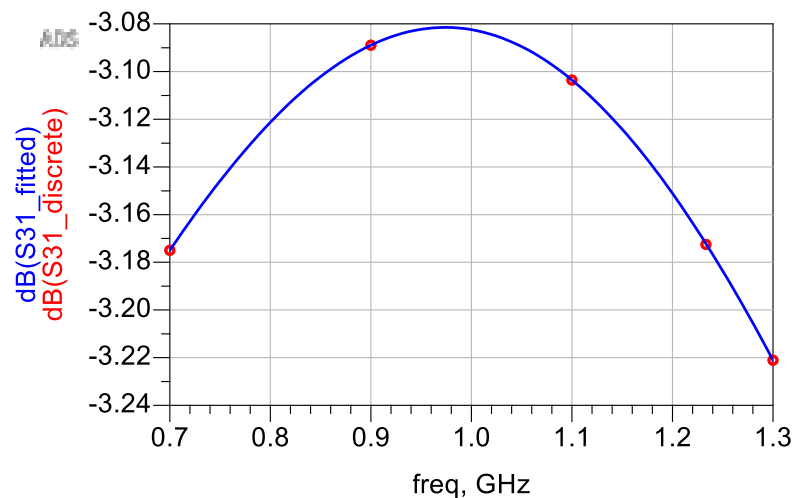
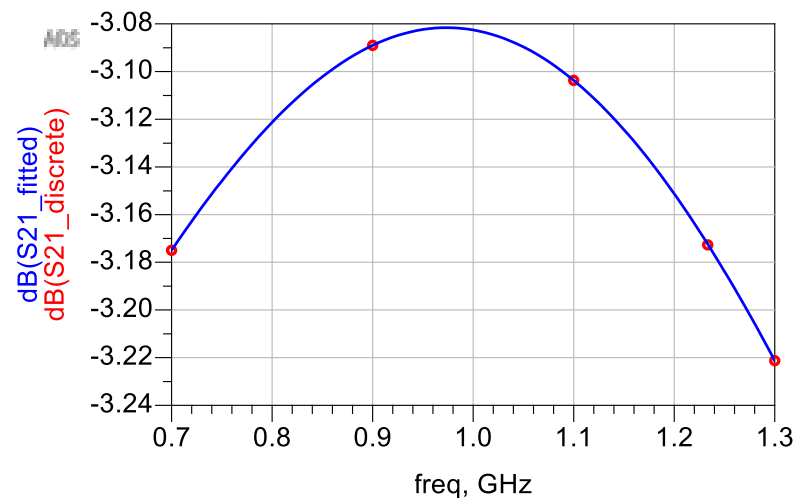
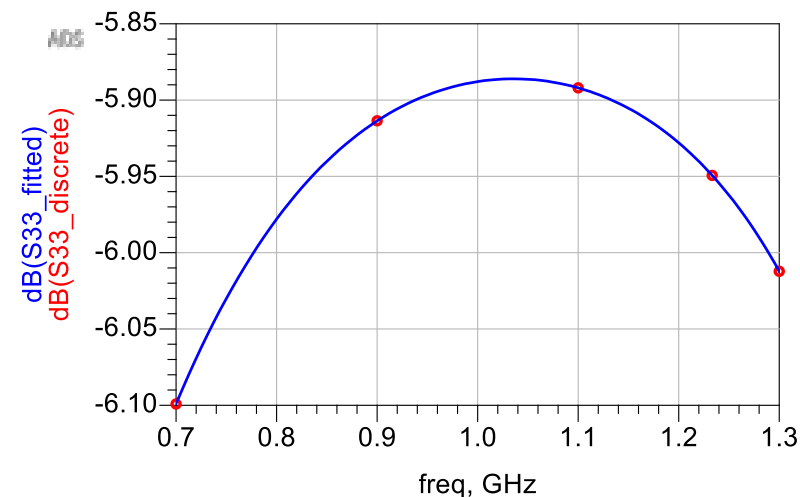
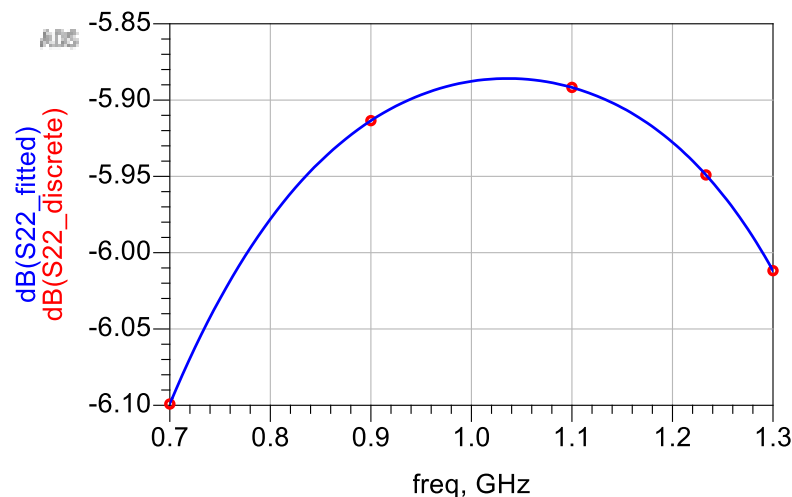
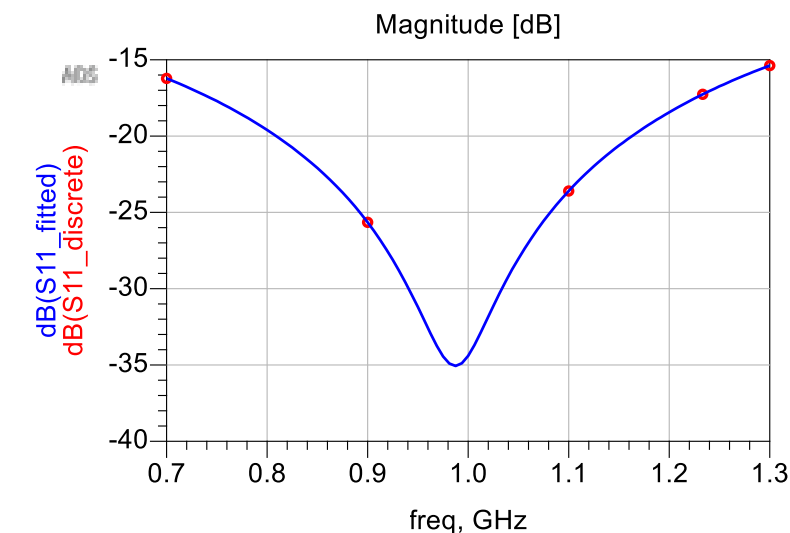


Type	Fstart	Fstop	Npts	Enabled
Adaptive	0.7 GHz	1.3 GHz	20 (max)	√



版图仿真结果, 结果并不好, 为什么?

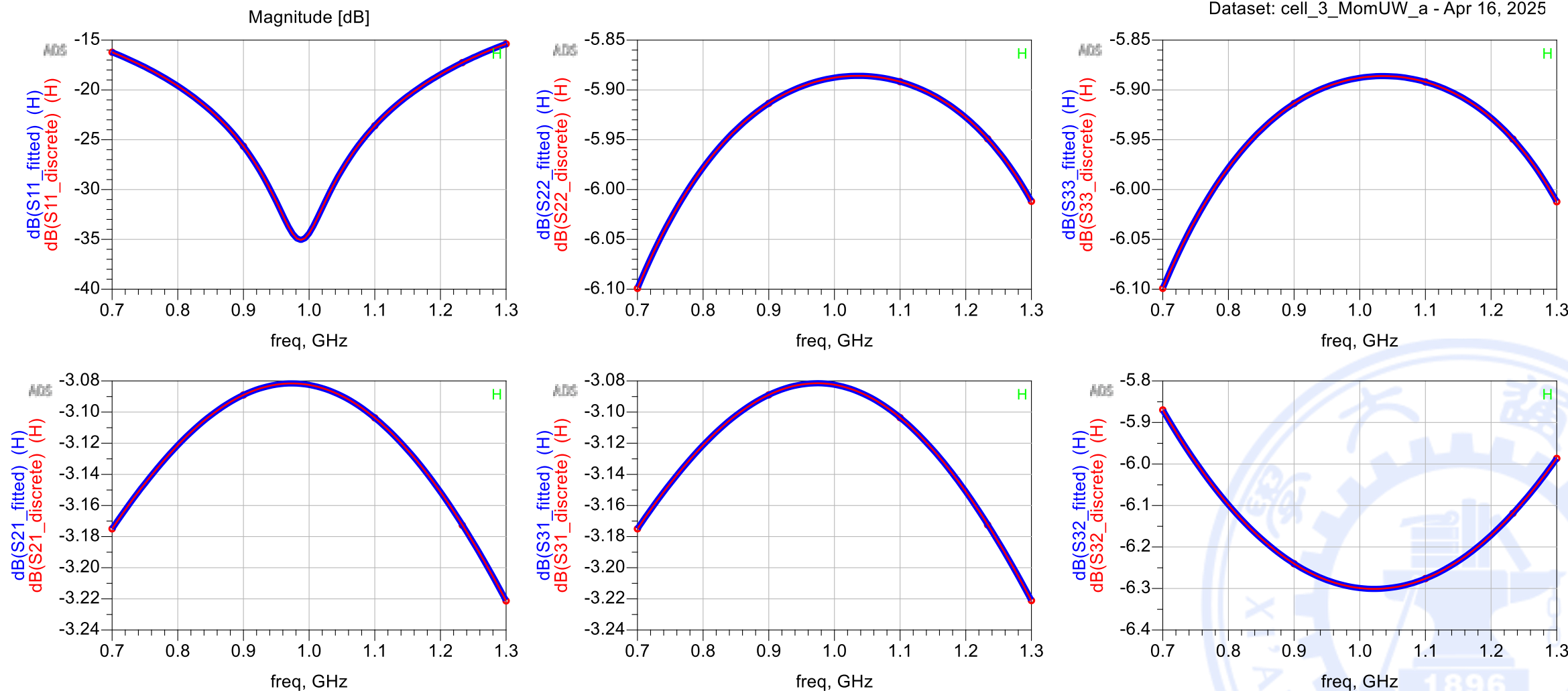
Dataset: cell_3_MomUW_a - Apr 16, 2025





9. 数据窗口，菜单【History】→【On】；

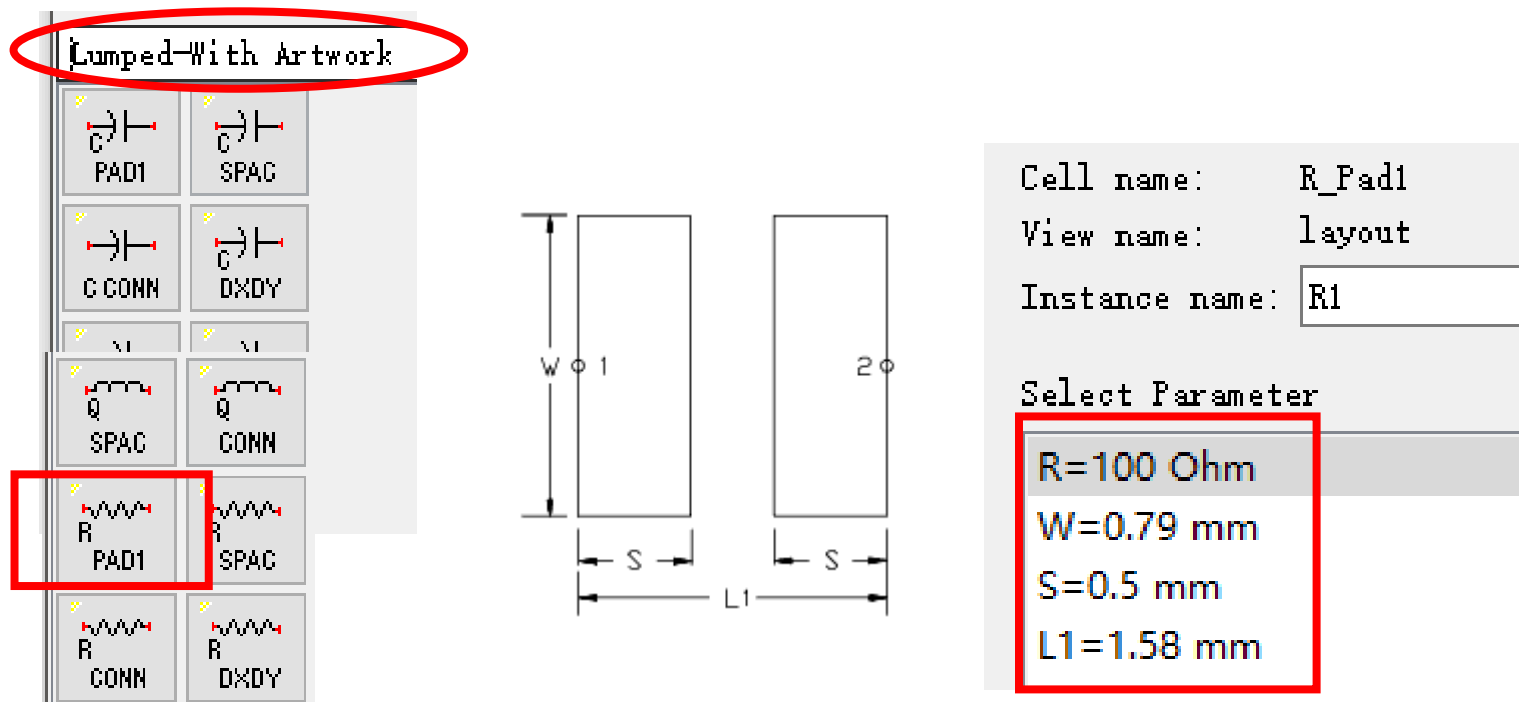
原理图→ TFR的Rs由50改为100，再次仿真。结果没有任何变化，说明电阻无效。在版图中无法直接仿真电阻。





方法1：版图中添加端口代替电阻

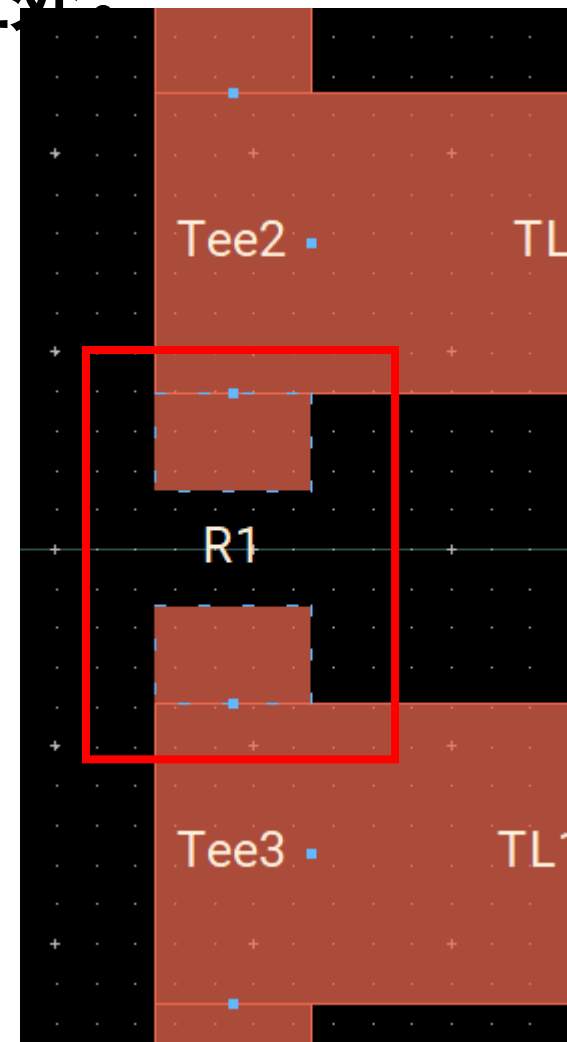
10. 版图中，删掉版图中电阻，
加入新电阻R_Pad1，设置参数，放在原电阻处。



R_Pad1的参数说明：

R在版图中只是表征隔离电阻为100欧姆，实际在版图仿真中不起任何作用，可以设置任意值。

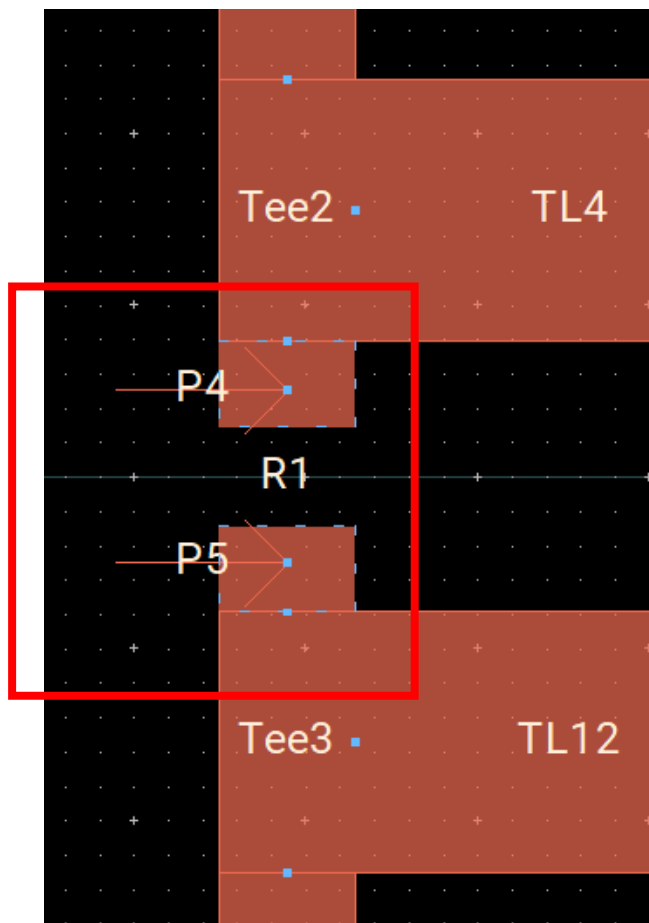
W，S，L1需根据所用电阻焊盘实际情况设定。





11. R_Pad1上加端口P4, P5, 用来替代隔离电阻。

- 菜单【EM】→【Port Editor】，把⊕ P5 拖到端口4的⊖ Gnd，并修改端口阻抗为100欧姆。重新仿真。



S-parameter Ports				
Number	Gnd Layer	Name	Feed Type	Ref Impedance
1	<Implicit>	P1	Auto	50 + 0i
2	<Implicit>	P2	Auto	50 + 0i
3	<Implicit>	P3	Auto	50 + 0i
4	<Implicit>	P4	Auto	50 + 0i
5	<Implicit>	P5	Auto	50 + 0i

Layout Pins			
Name	Layer	Net	Connected
P1	cond	P1	1(+)
P2	cond	P2	2(+)
P3	cond	P3	3(+)
P4	cond	P4	4(+)
P5	cond	P5	5(+)

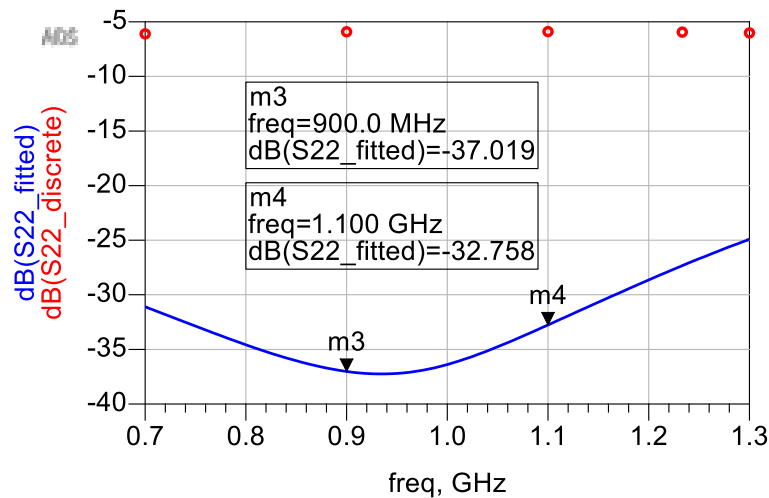
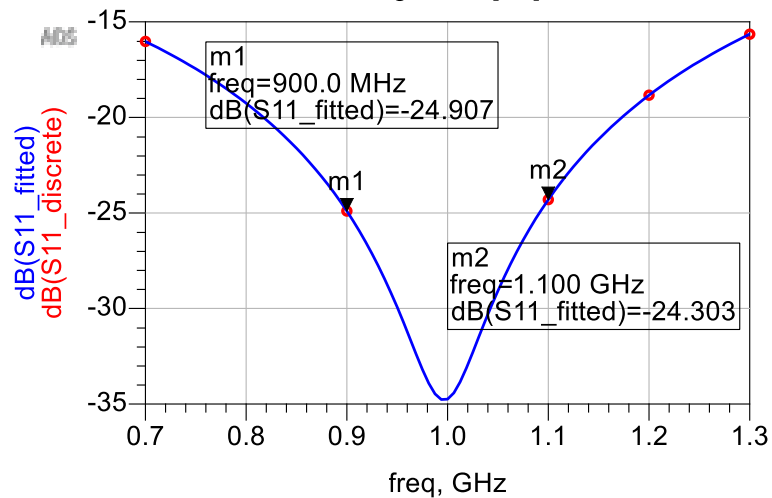
Number	Gnd Layer	Name	Feed Type	Ref Impedance
1	<Implicit>	P1	Auto	50 + 0i
2	<Implicit>	P2	Auto	50 + 0i
3	<Implicit>	P3	Auto	50 + 0i
4	N/A	P4	Auto	100 + 0i

Layout Pins			
Name	Layer	Net	Connected
P1	cond	P1	1(+)
P2	cond	P2	2(+)
P3	cond	P3	3(+)
P4	cond	P4	4(+)
P5	cond	P5	4(-)

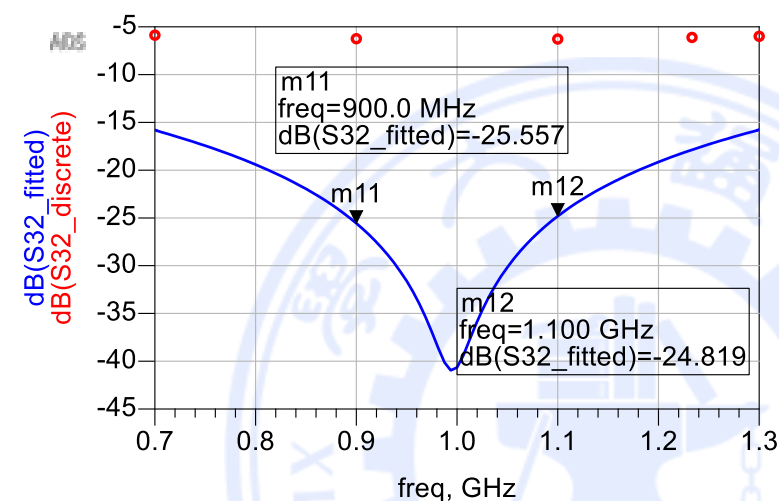
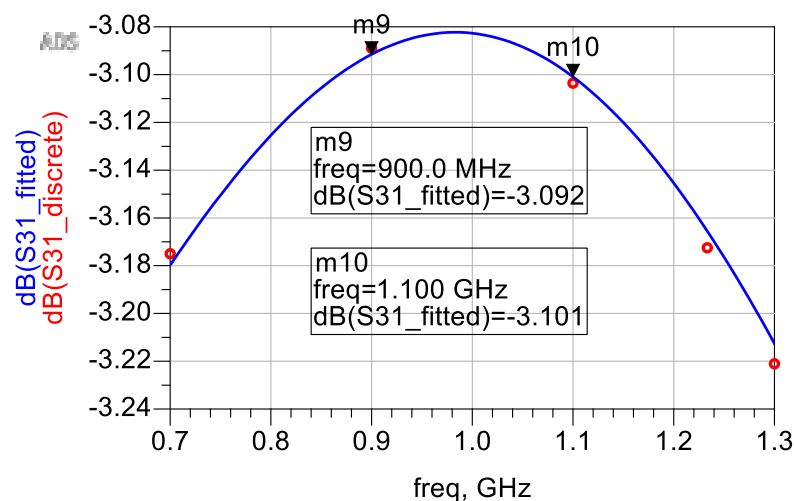
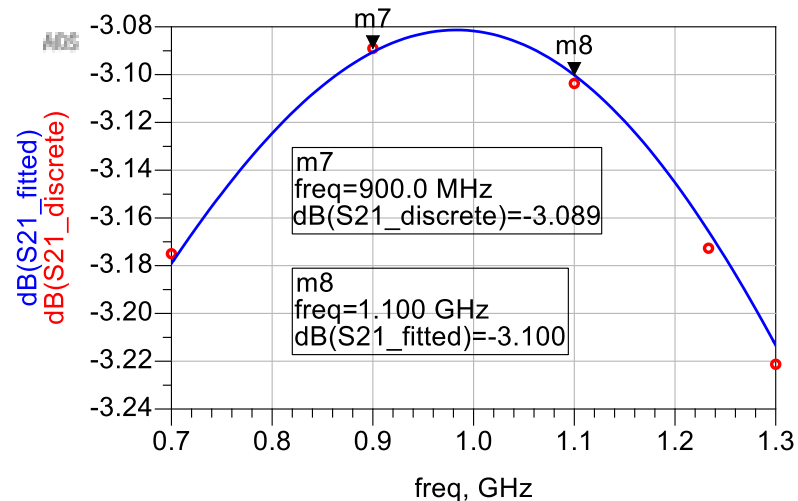
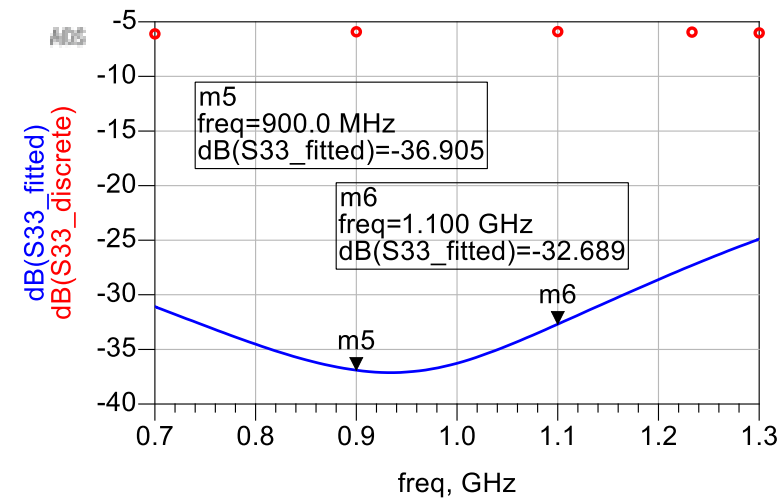


版图仿真结果，基本满足指标要求

Magnitude [dB]



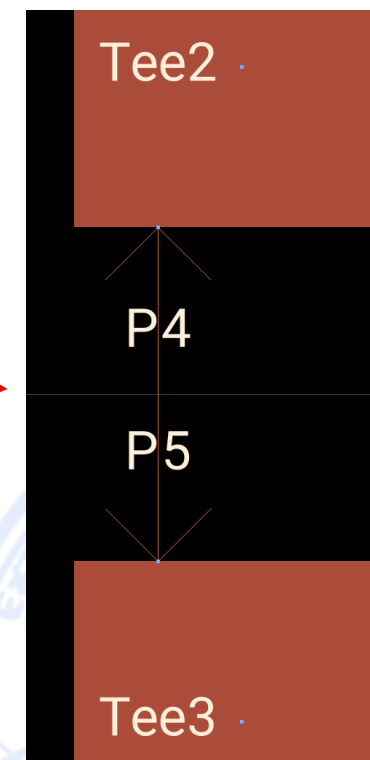
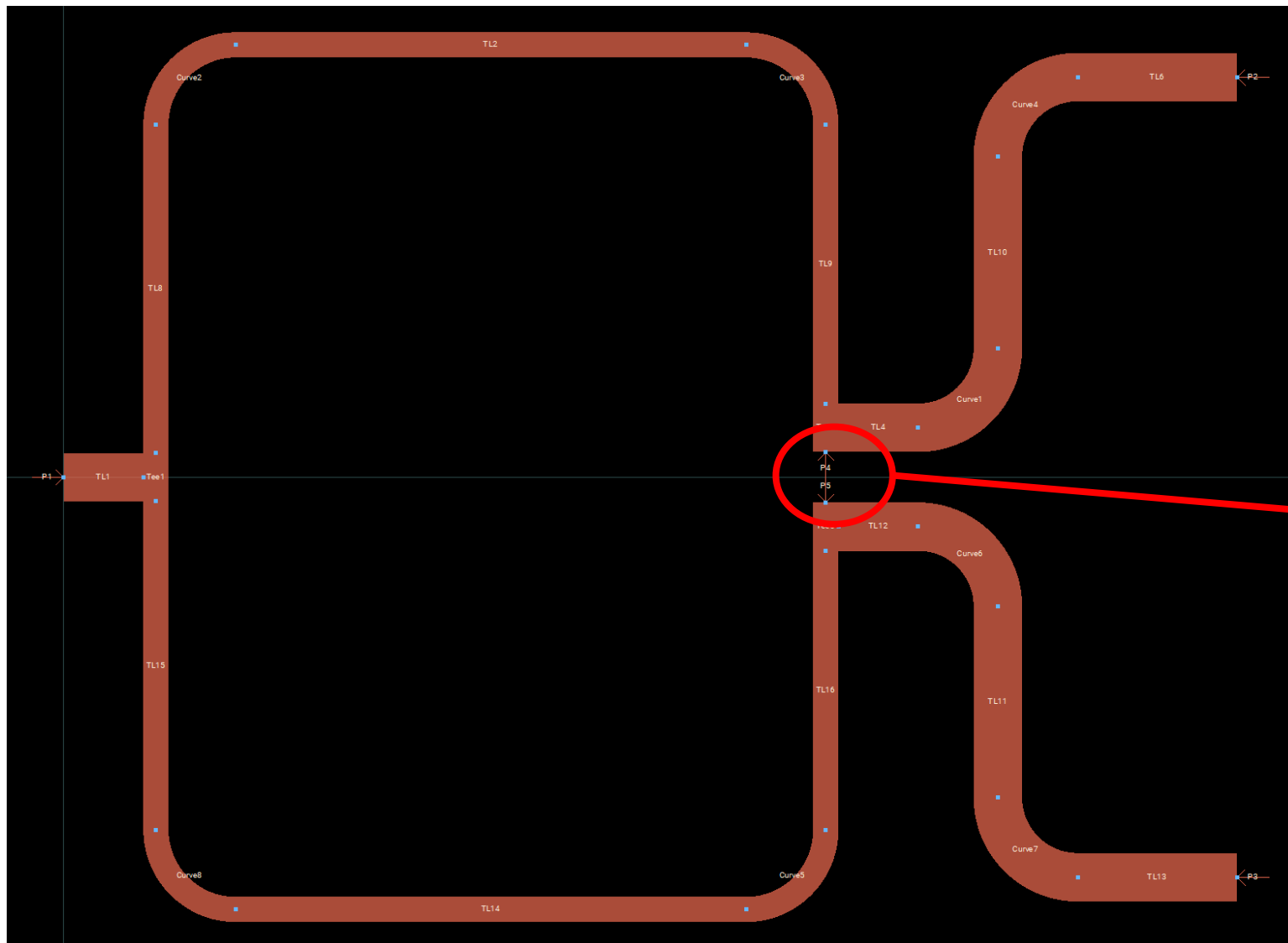
Dataset: cell_3_MomUW_a - Apr 16, 2025





方法2：版图-原理图联合仿真

接步骤8，版图中，删掉版图中电阻，添加端口

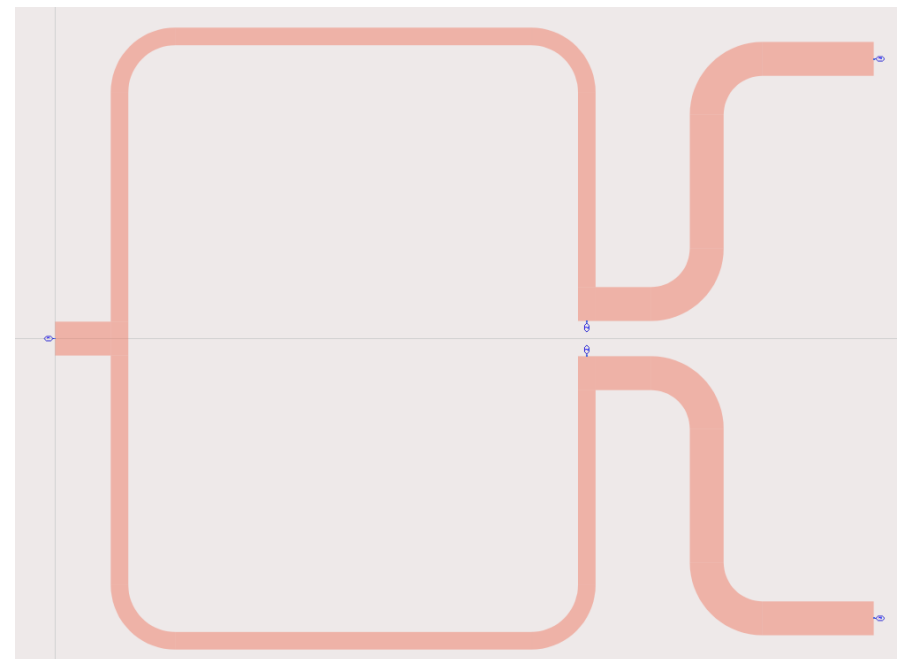
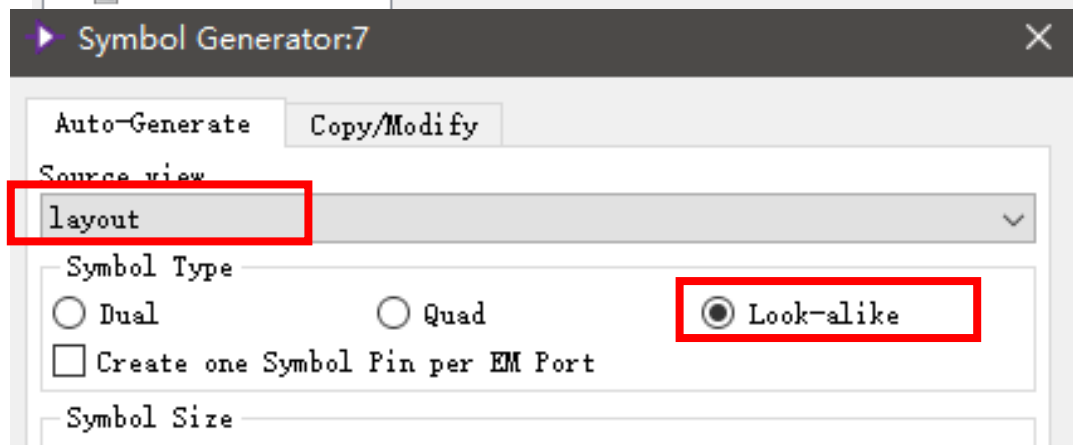
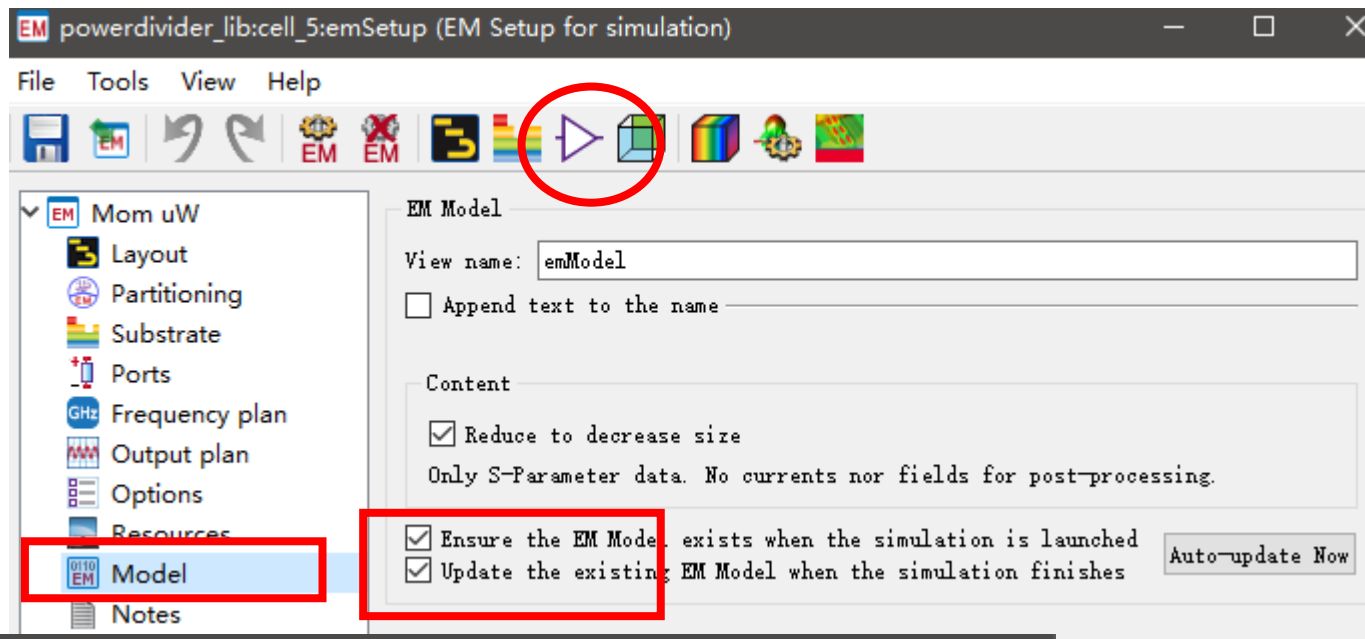




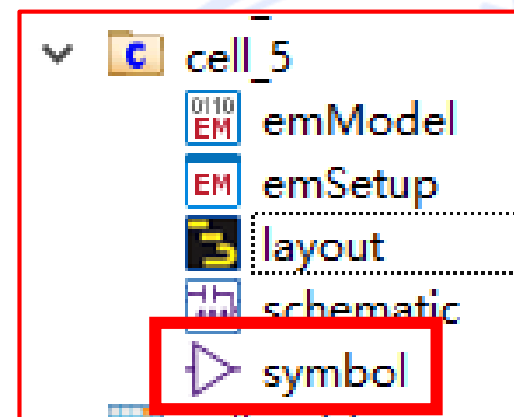
➤ 菜单【EM】→【Simulation Setup】或者 ，打开仿真设置窗口

➤ 【Model】，选中Ensure

➤ 单击  打开符号编辑器，生成symbol。



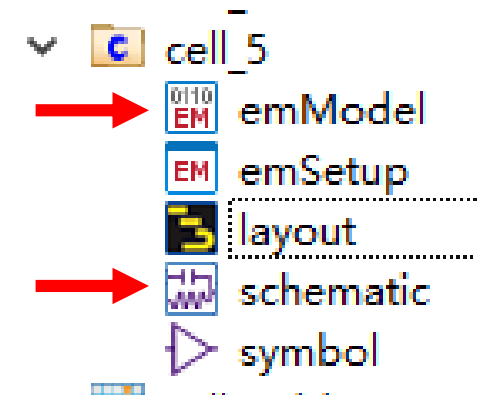
在对应的cell
下可以看到
该symbol





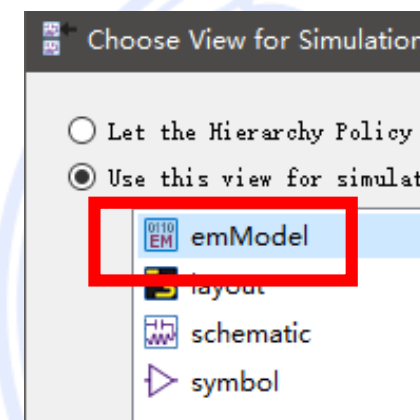
➤ 新建原理图，拖入symbol，把前面原理图中的SP控件，3个Term，TFR电阻，Msub拷贝过来

有两个视图



单击

选中emModel，然后仿真





作业：功分器仿真设计

设计一微带结构的威尔金森功分器，指标要求：

中心频率：单号同学：1.0 GHz+学号后两位，带宽：200MHz

双号同学：2.0 GHz+学号后两位，带宽：300MHz

如：学号11， $1.0+0.11=1.11\text{GHz}$ ，学号12， $2.0+0.12=2.12\text{GHz}$

（如果后两位为0开头，则取最后一位，如学号03， $1.0+0.3=1.3\text{GHz}$ ）

输出端口功率比：1: 1

频带内输入端口的回波损耗： $S_{11} < -20\text{dB}$

隔离度： $S_{32} < -20\text{dB}$

频带内插入损耗： $S_{21} > -3.1\text{dB}$, $S_{31} > -3.1\text{dB}$

板材选用Rogers_RT_Duroid5880

基板厚度：0.508mm

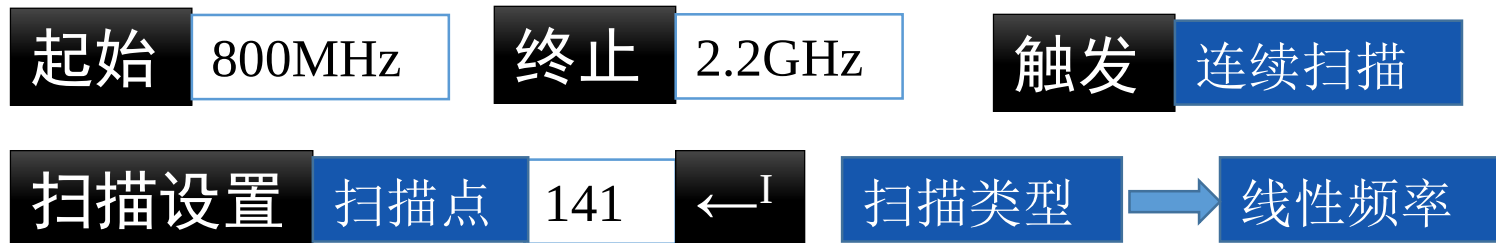
覆铜厚度：0.035mm



三、功分器的测量

测功分器在800MHz-2200MHz的回损、插损和隔离度

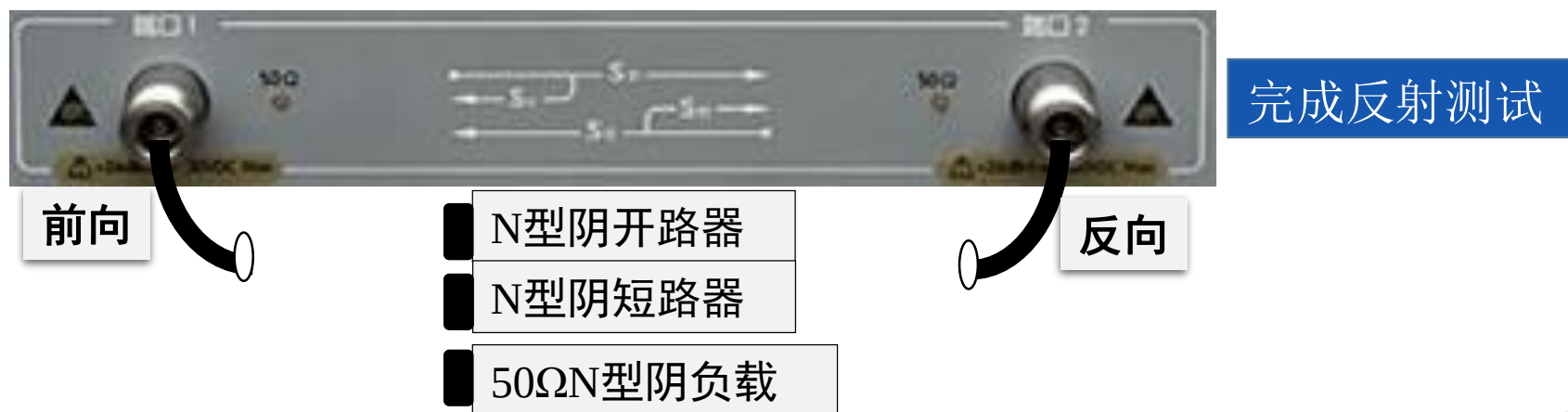
① 设置激励参数



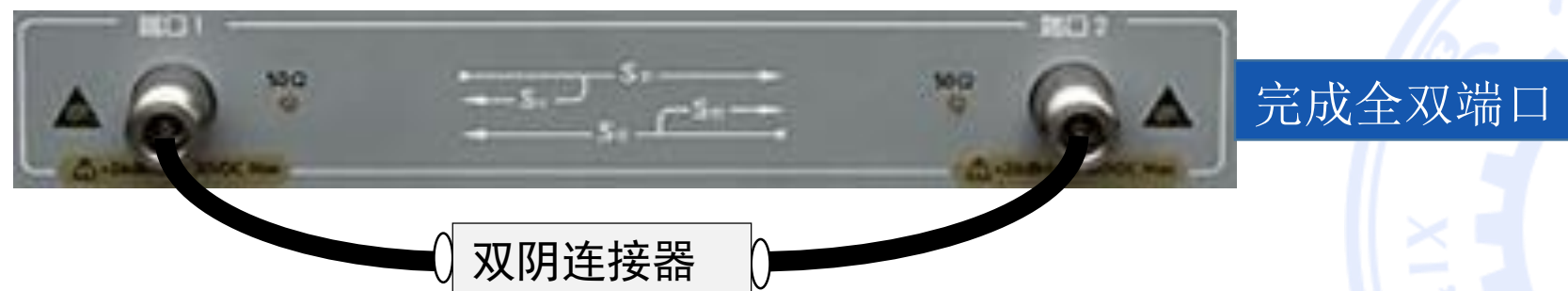


② 进行测量校准

校准 → 机械校准 → 全双端口 (SOLT) → 反射 按图连接
分别进行前向和反向开路、短路、匹配负载校准



按图连接，进行直通校准 传输 → 直通 完成传输测试

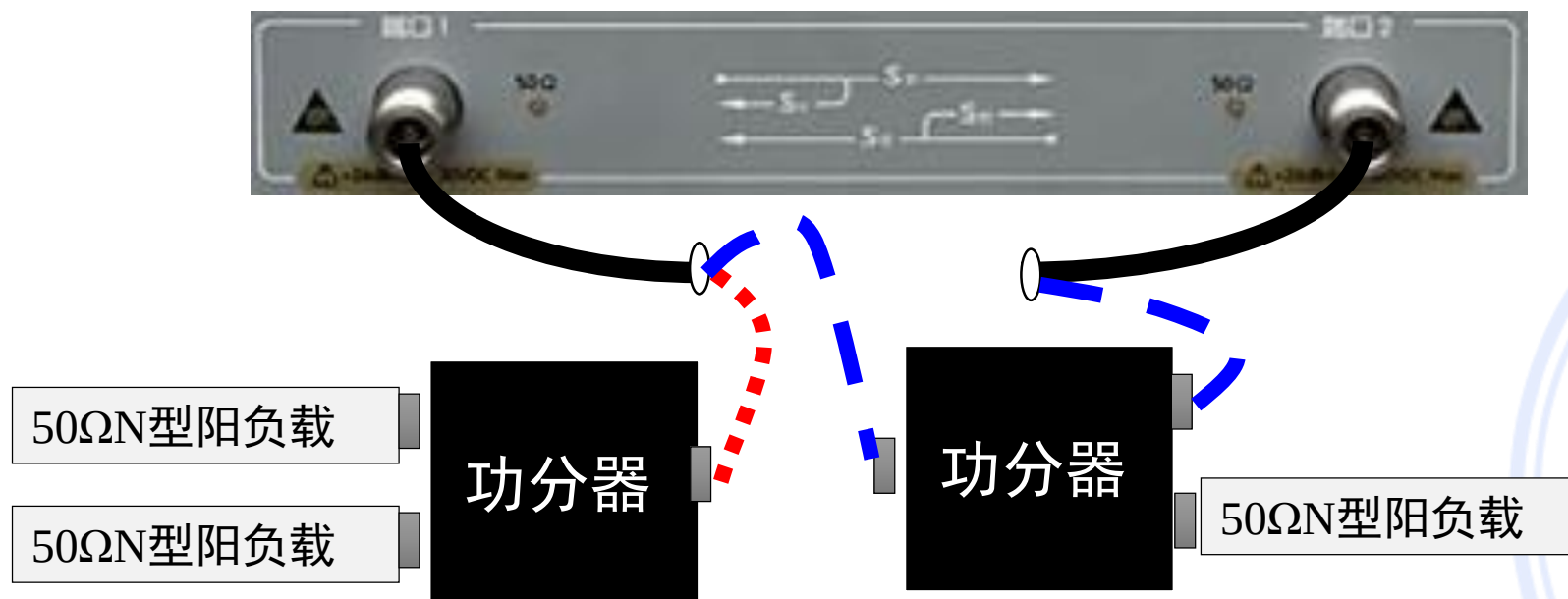




③ 测量，记录结果

测量	S11	S21
格式	驻波 + 对数幅度	对数幅度 + 相位

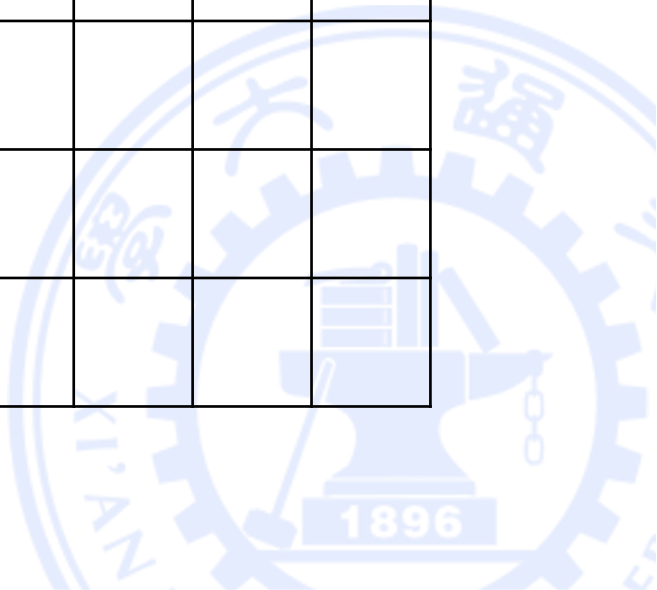
光标 用调节旋钮分别记录**驻波和回损**（输入端）；**插损和相移**（支路1、支路2）和**隔离度**（支路1与支路2，应该如何连接？），记录（15组），并照相记录波形。





功分器参数测量记录表

频率(GHz)															
回损(dB)															
驻波															
支路1 插损(dB)															
支路1 相移															
支路2 插损(dB)															
支路2 相移															
隔离度(dB)															





四、思考题

1. 给出四种用ADS微带线计算工具LinCalc可计算、设计的微带线
2. 微带圆弯头与最优弯头有什么区别？
3. 威尔金森功分器中的电阻 R 理论取值是多大？起什么作用？
4. 比较电阻功分器与威尔金森功分器的特点。



五、本次实验报告格式

一、实验目的

二、威尔金森功分器

- (1) 叙述任意分配比威尔金森功分器**原理**；
- (2) 写明所仿功分器的中心频率；
- (3) 给出**原理图**和**仿真结果**，弯头必须用**最优弯头**（如果效果不好需优化）；
- (4) 给出**版图**和**仿真结果**。

三、功分器测试结果及分析

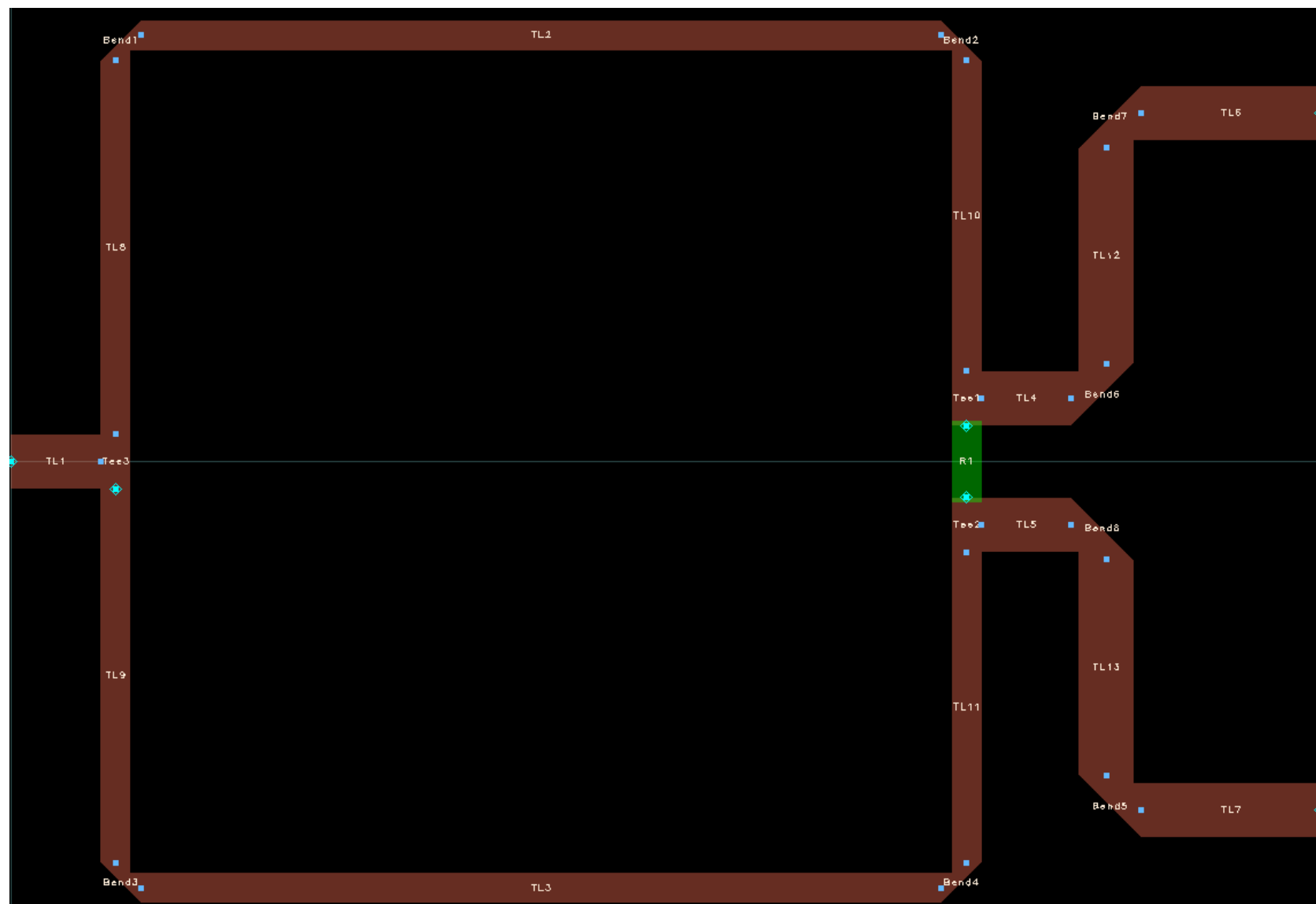
- (1) 数据表格；
- (2) 可以体现被测件特性的照片2张（1张支路1插损，1张支路1相移）；
- (3) 数据曲线（**回损**、**插损**（**支路1**、**支路2**）、**隔离度**）；
- (4) 结果分析。

四、实验体会和建议

五、思考题



最优弯头





西安交通大学
XI'AN JIAOTONG UNIVERSITY

谢谢！

